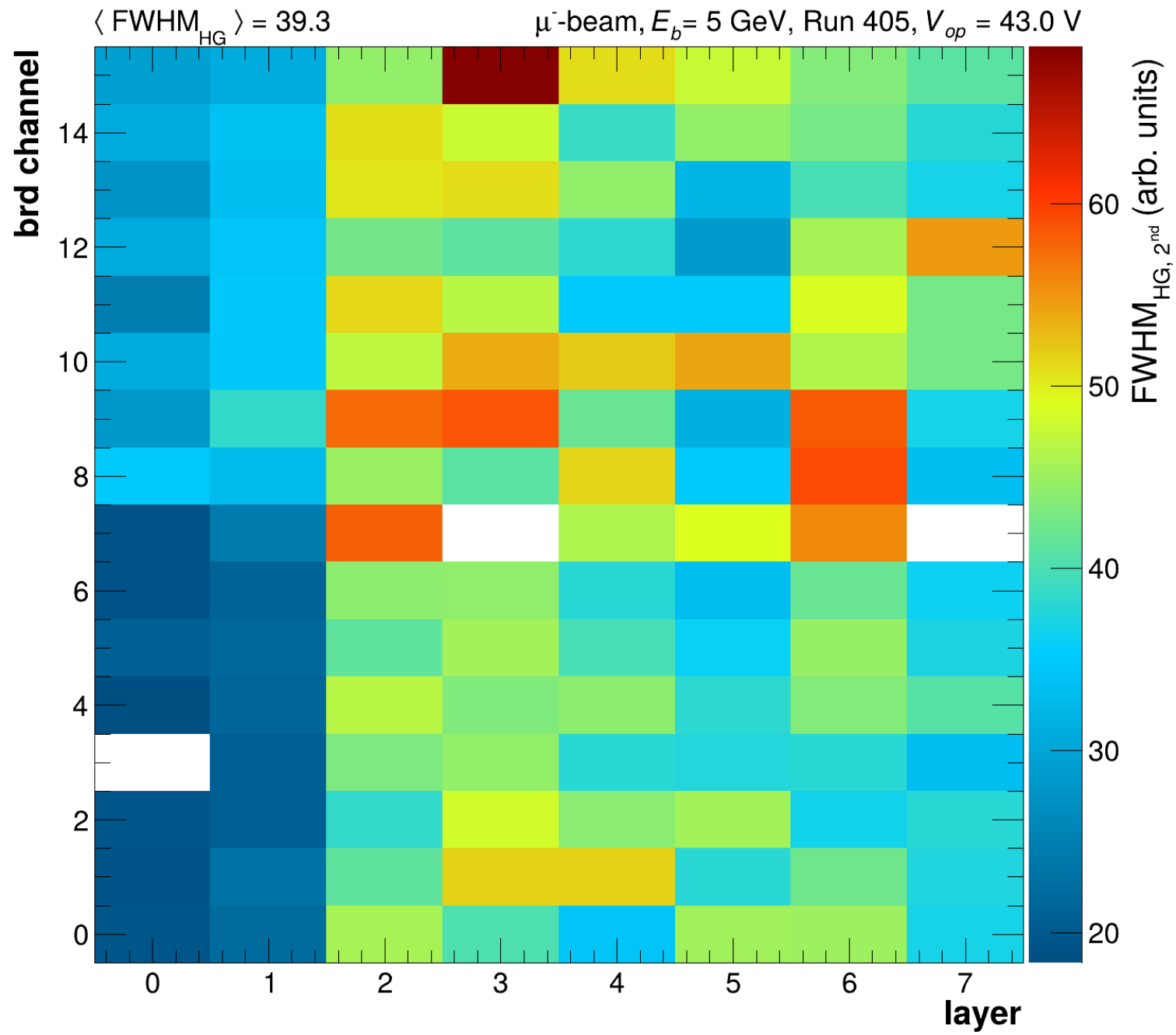
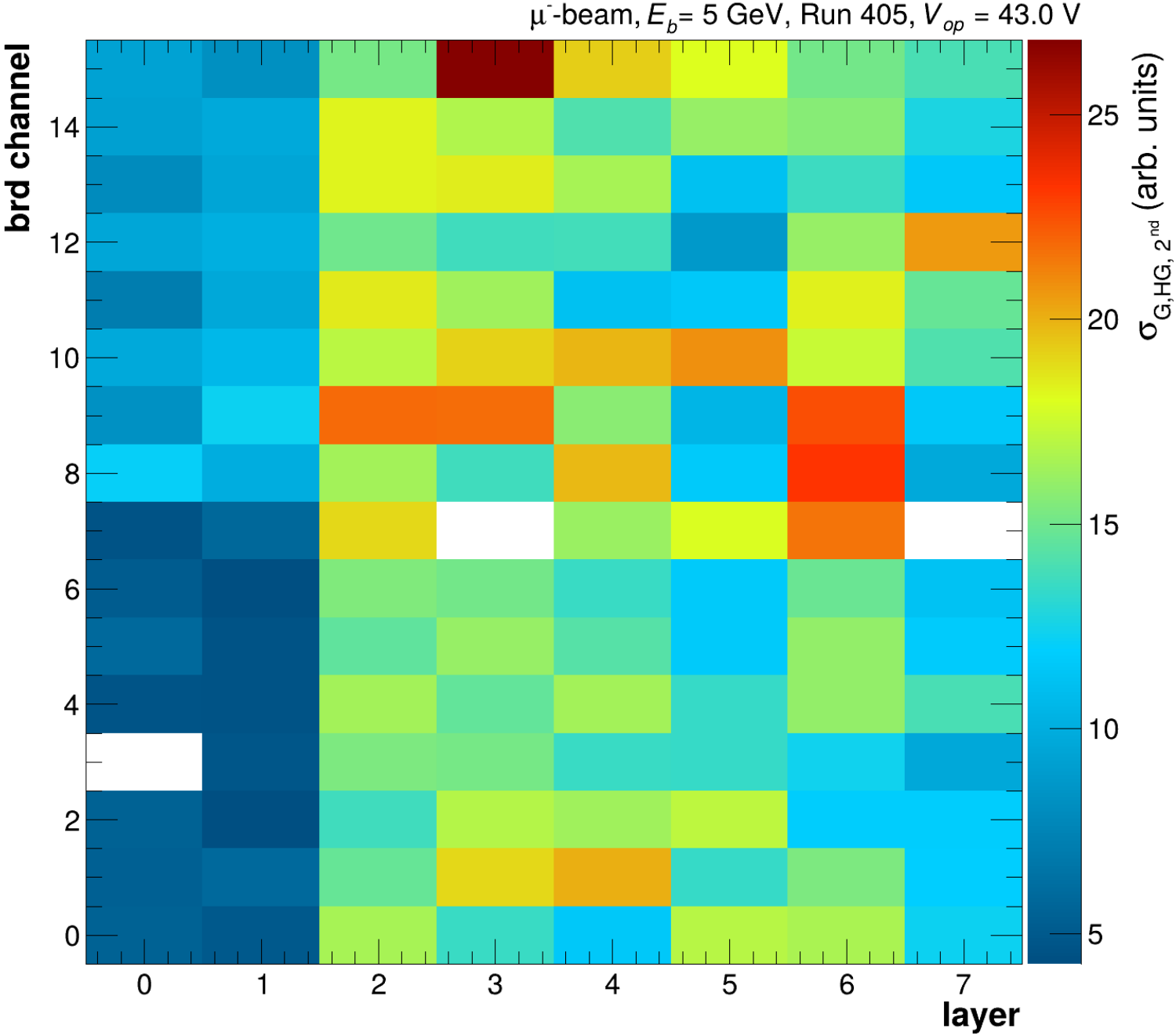
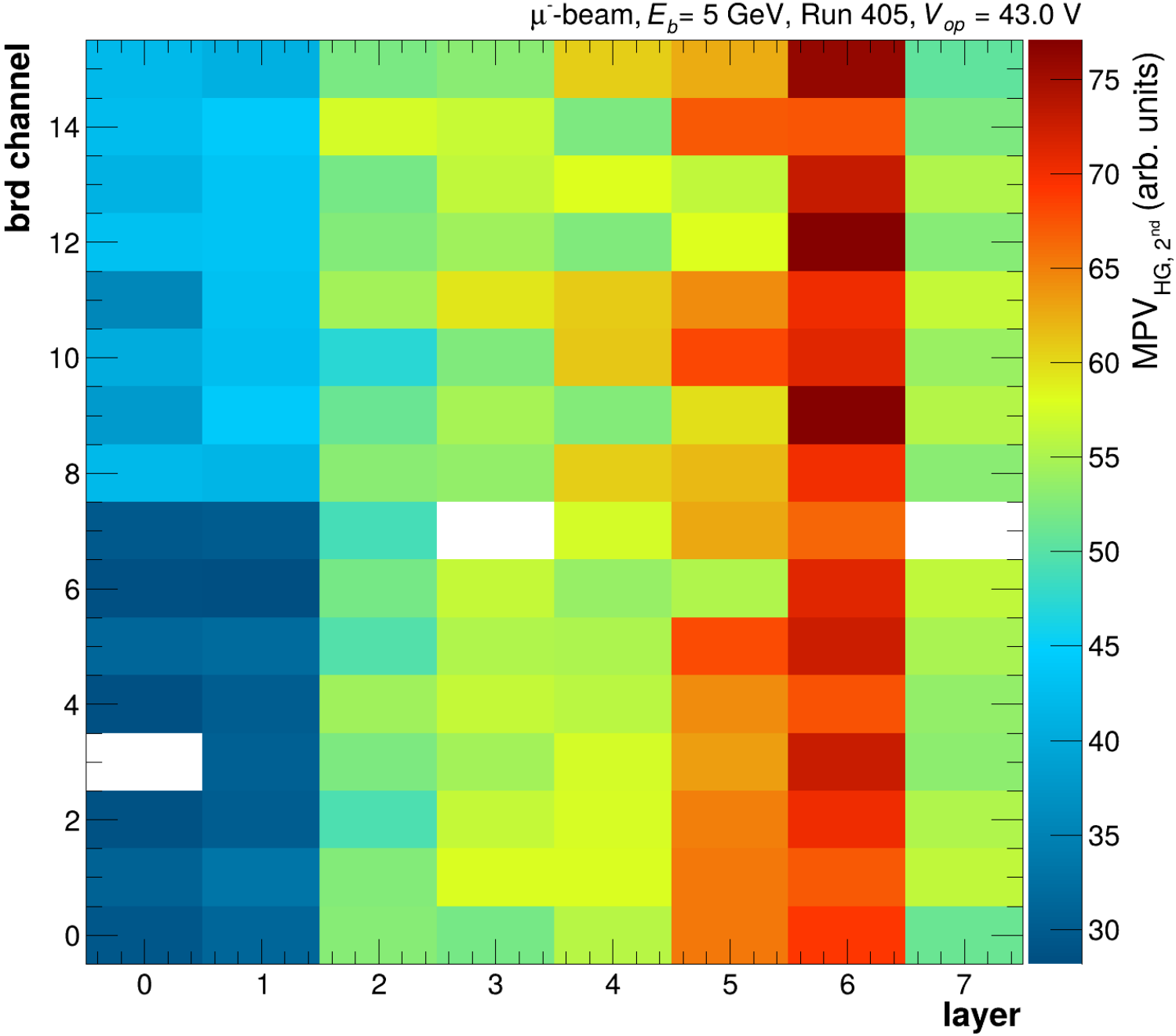
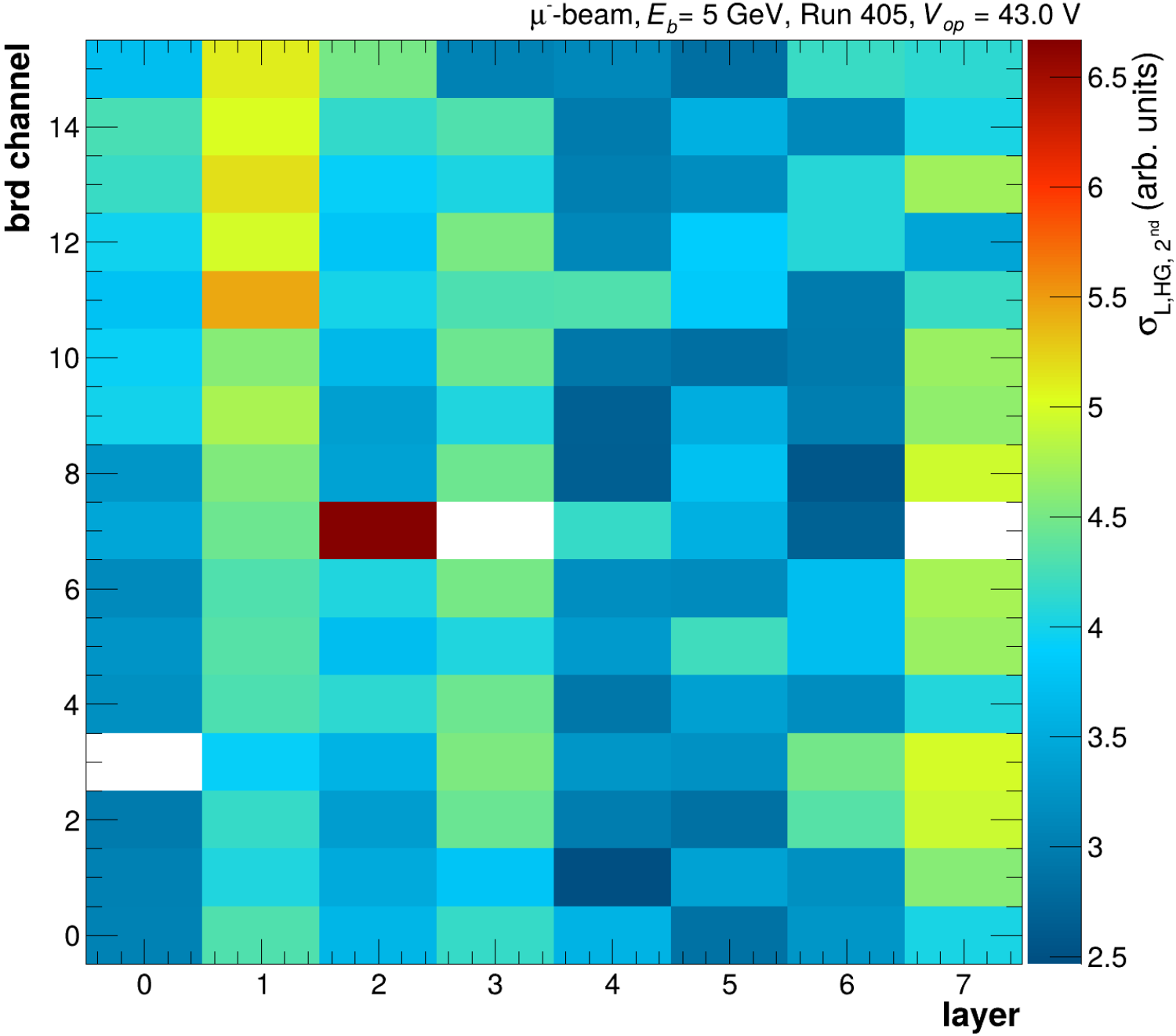
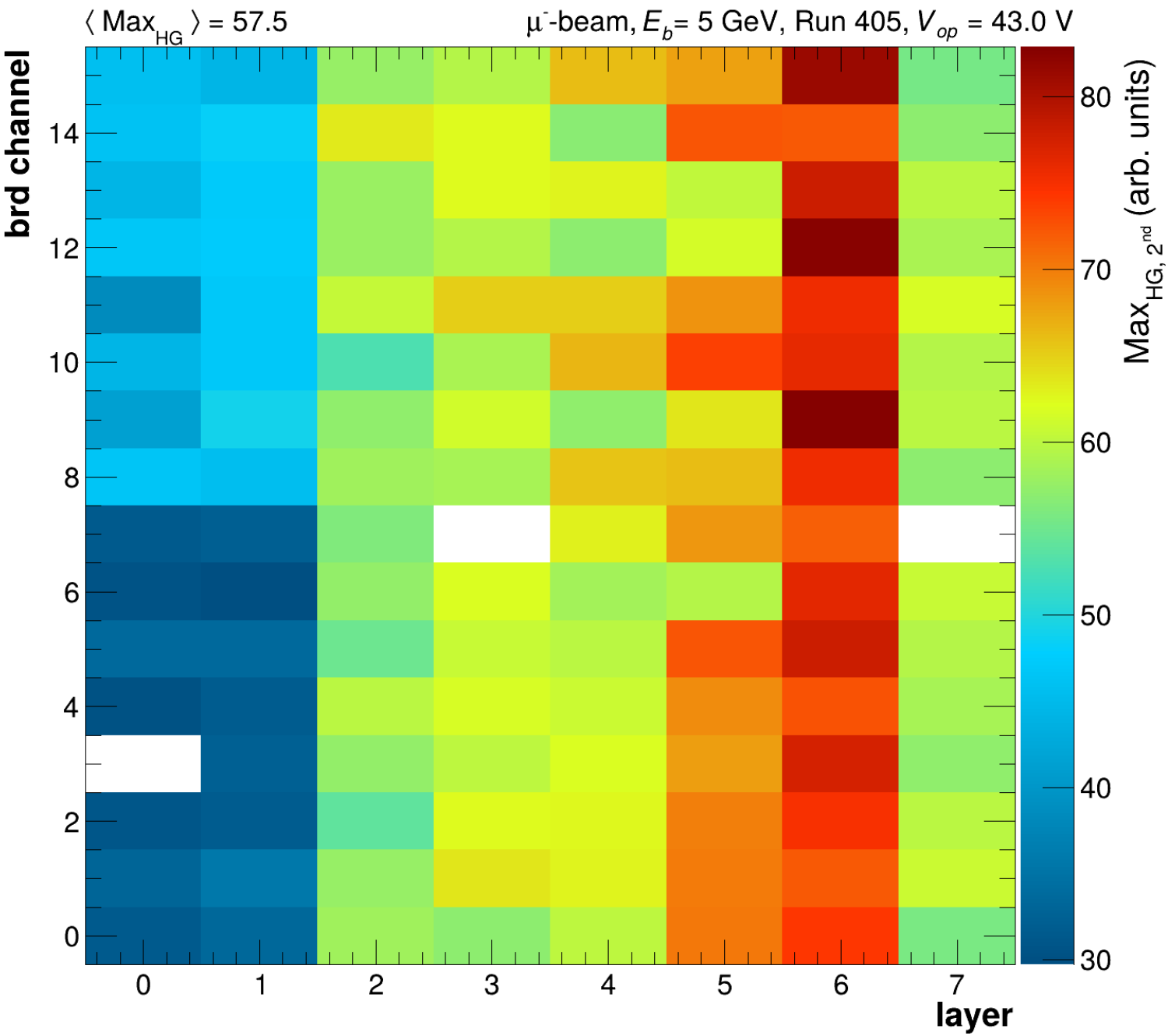
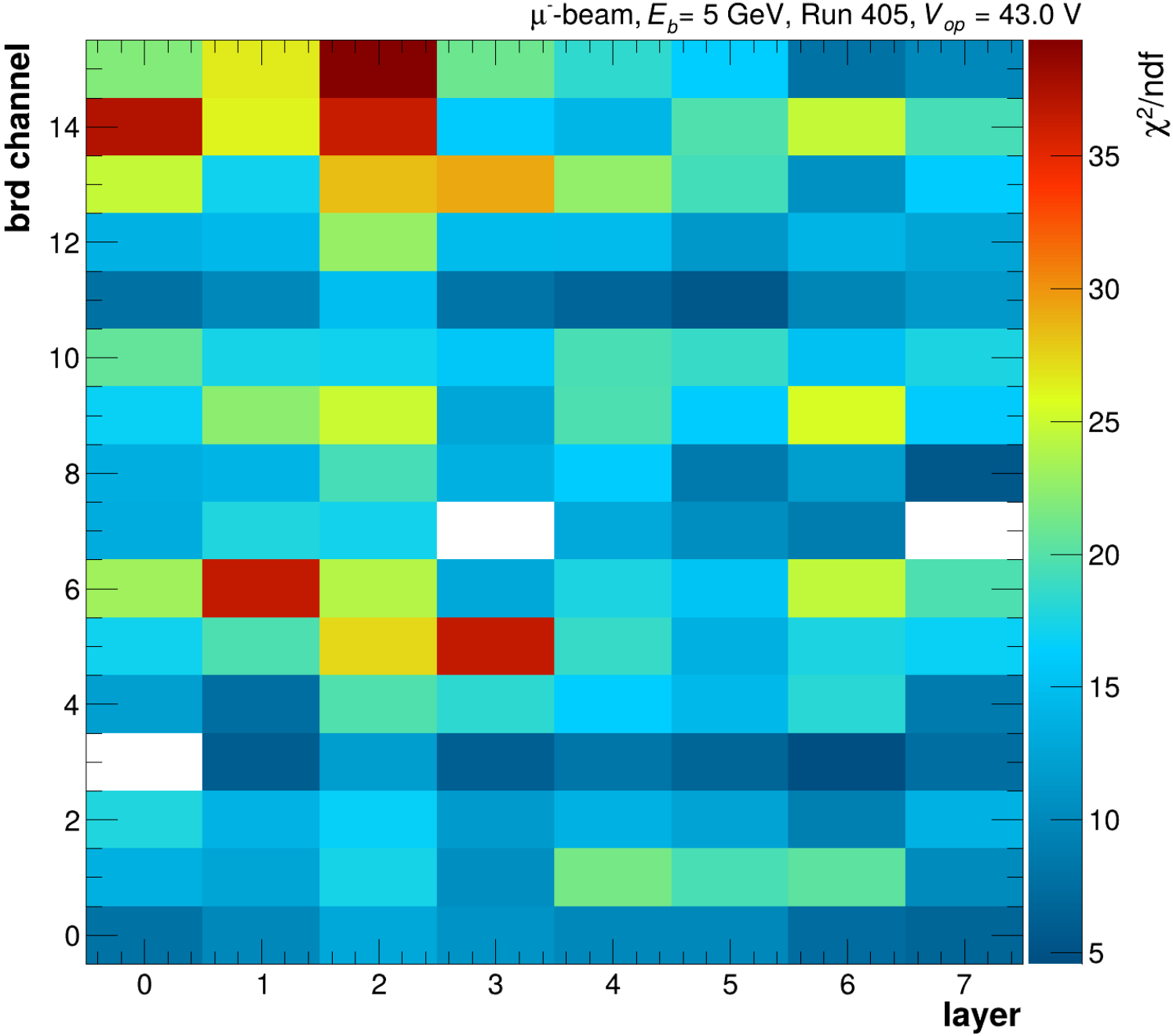

⟨ FWHM_HG ⟩ = 39.3
μ⁻-beam, E_b = 5 GeV, Run 405, V_op = 43.0 V
brd channel
layer
FWHM_HG, 2nd (arb. units)
0
2
4
6
8
10
12
14
1
3
5
7
20
30
40
50
60

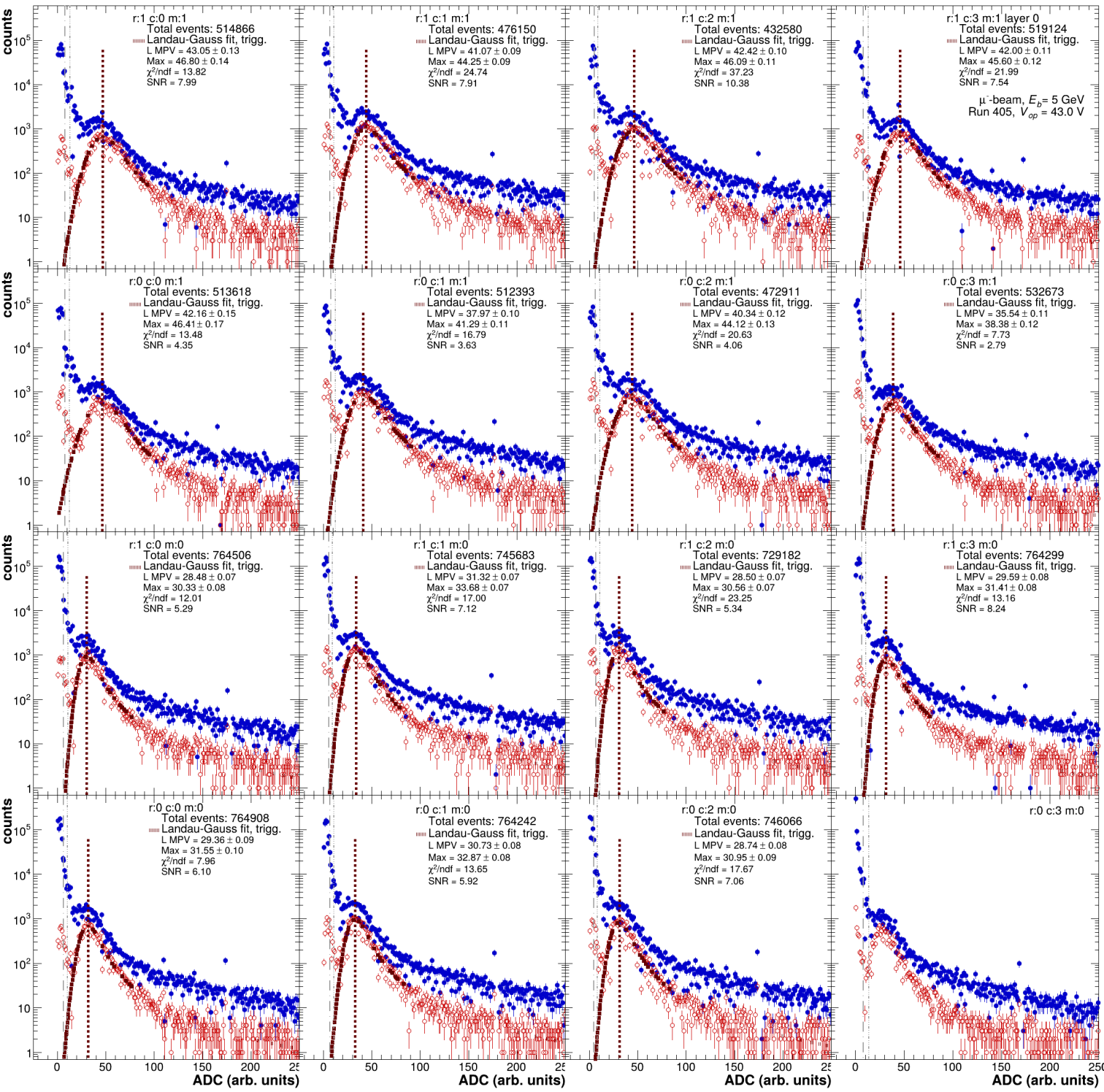

μ⁻-beam, E_b= 5 GeV, Run 405, V_{op} = 43.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
5
10
15
20
25
$\sigma_{G,HG,2^{nd}}$ (arb. units)

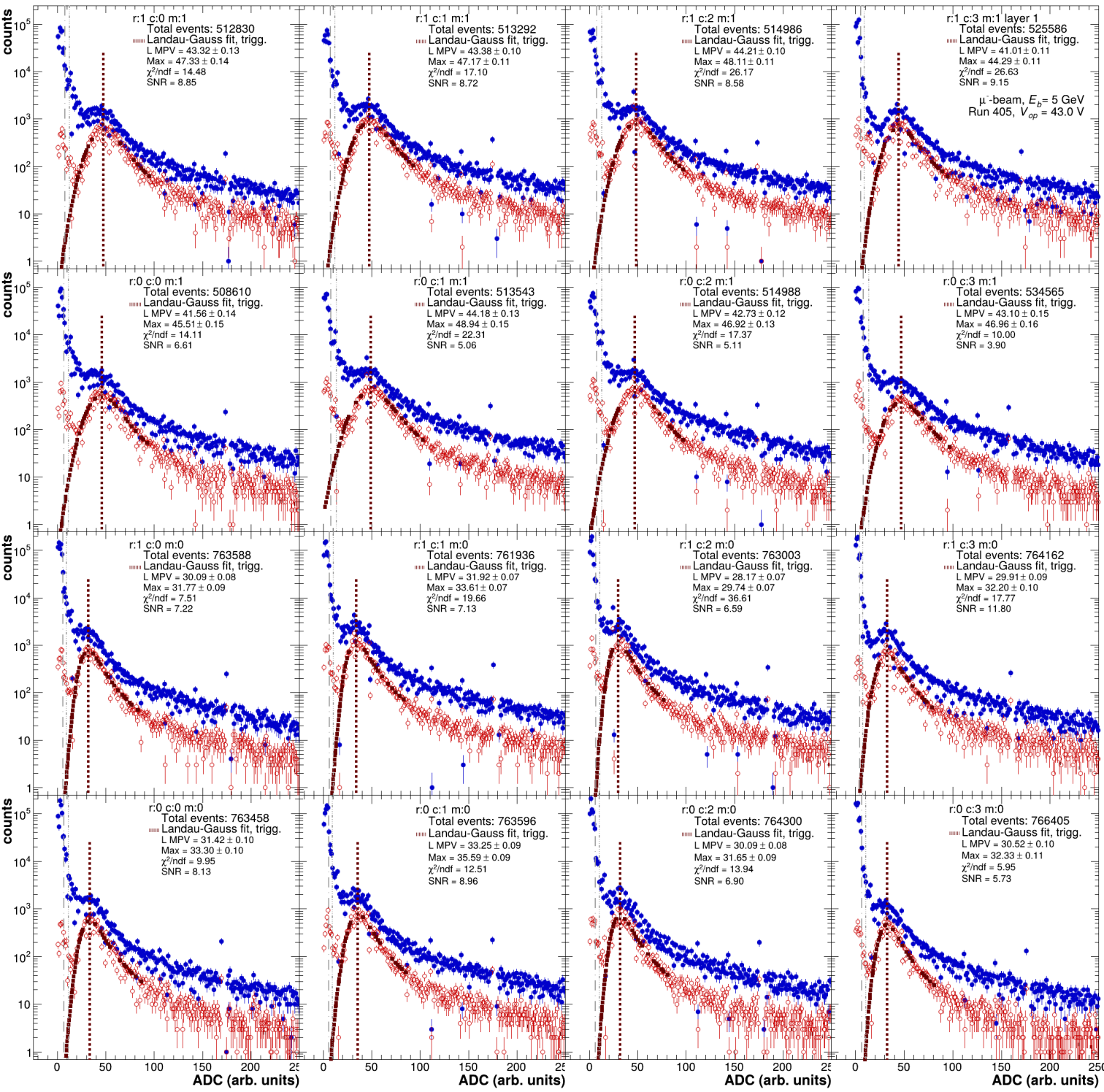

μ⁻-beam, E_b= 5 GeV, Run 405, V_{op} = 43.0 V
brd channel
layer
$MPV_{HG,\ 2^{nd}}$ (arb. units)
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
30
35
40
45
50
55
60
65
70
75

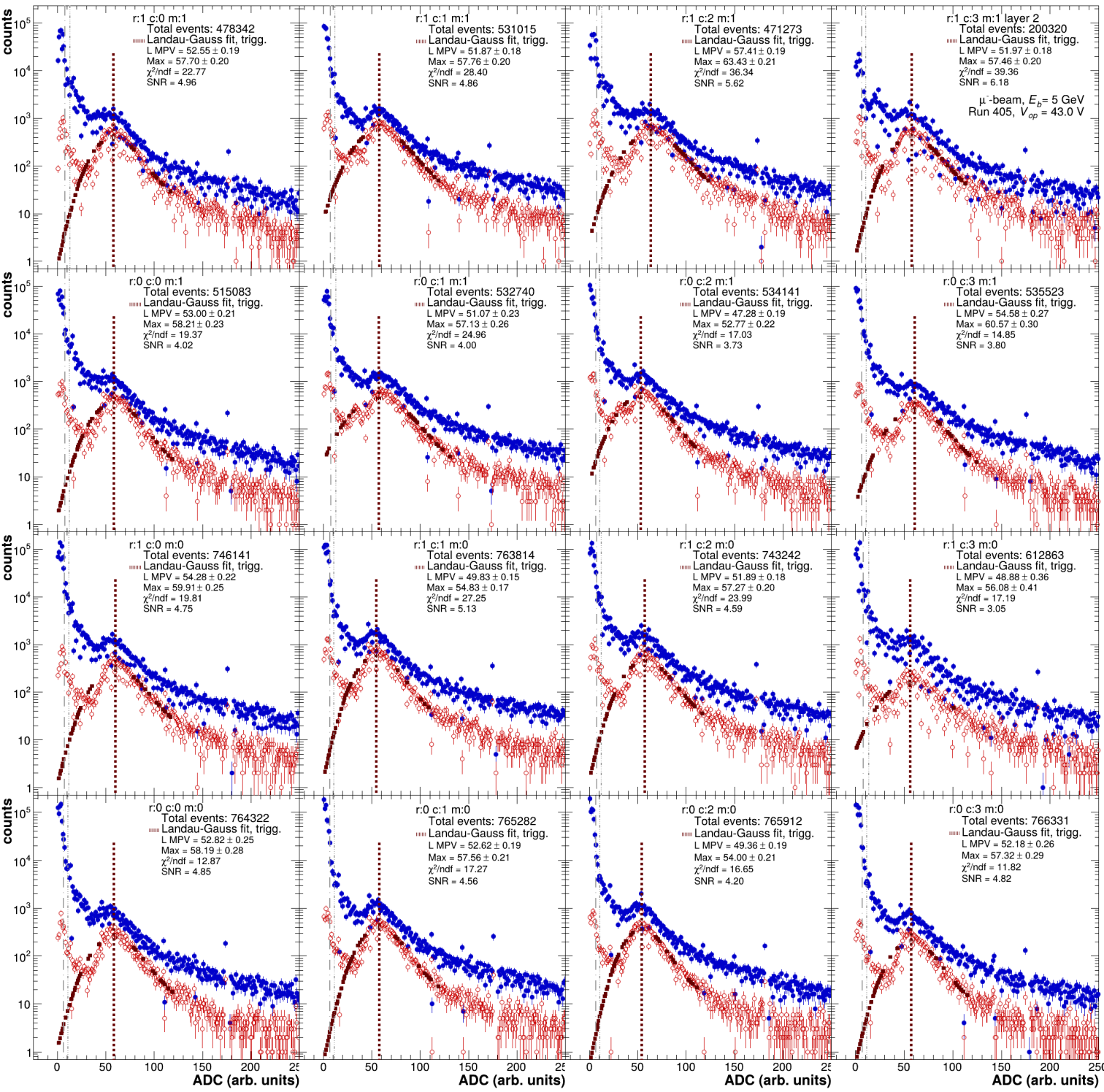

μ⁻-beam, E_b= 5 GeV, Run 405, V_{op} = 43.0 V
brd channel
layer
$\sigma_{L,HG,2^{nd}}$ (arb. units)
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
2.5
3
3.5
4
4.5
5
5.5
6
6.5

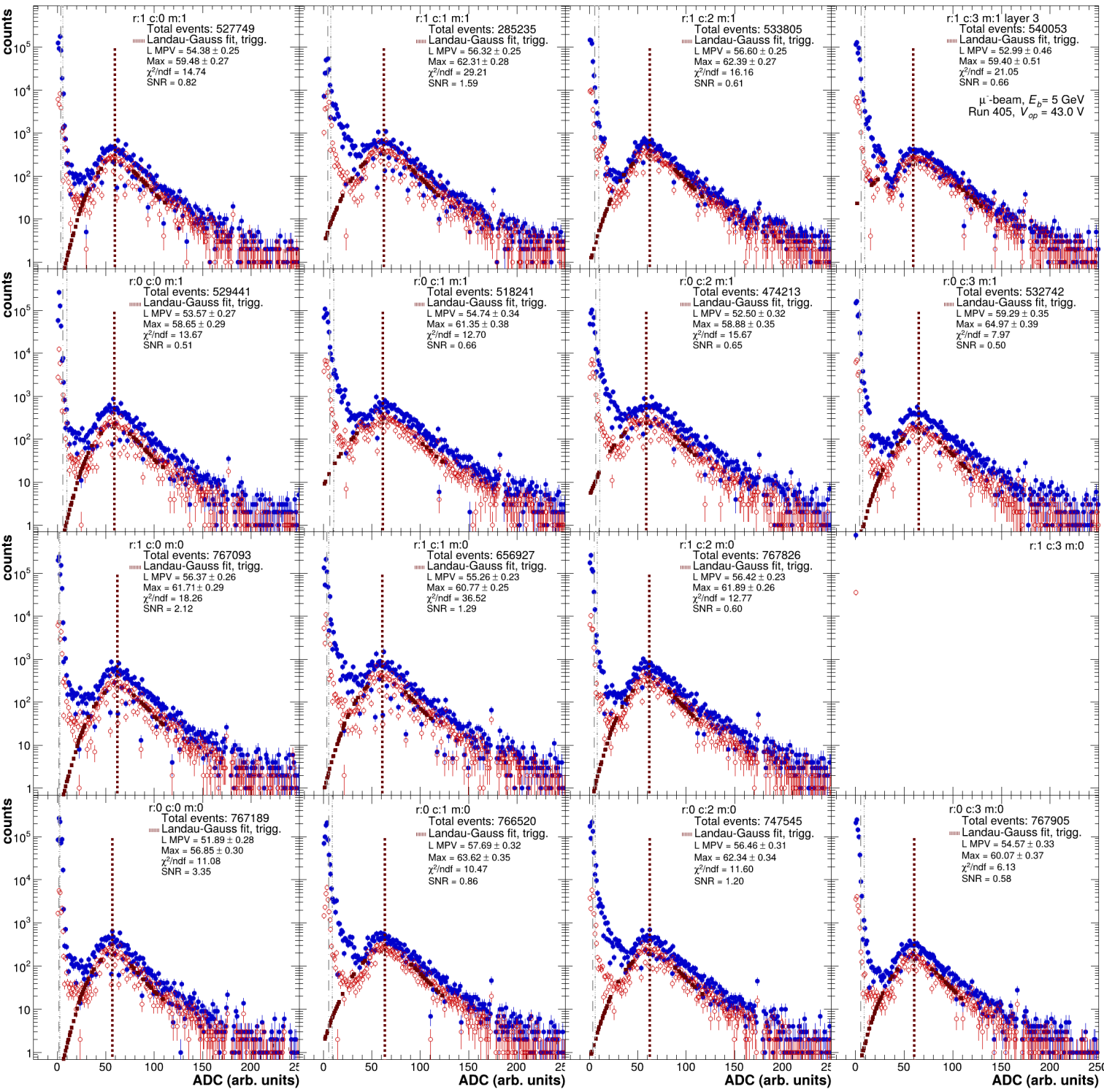

⟨ Max_HG ⟩ = 57.5
μ⁻-beam, E_b = 5 GeV, Run 405, V_op = 43.0 V
brd channel
layer
Max_HG, 2nd (arb. units)
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
30
40
50
60
70
80

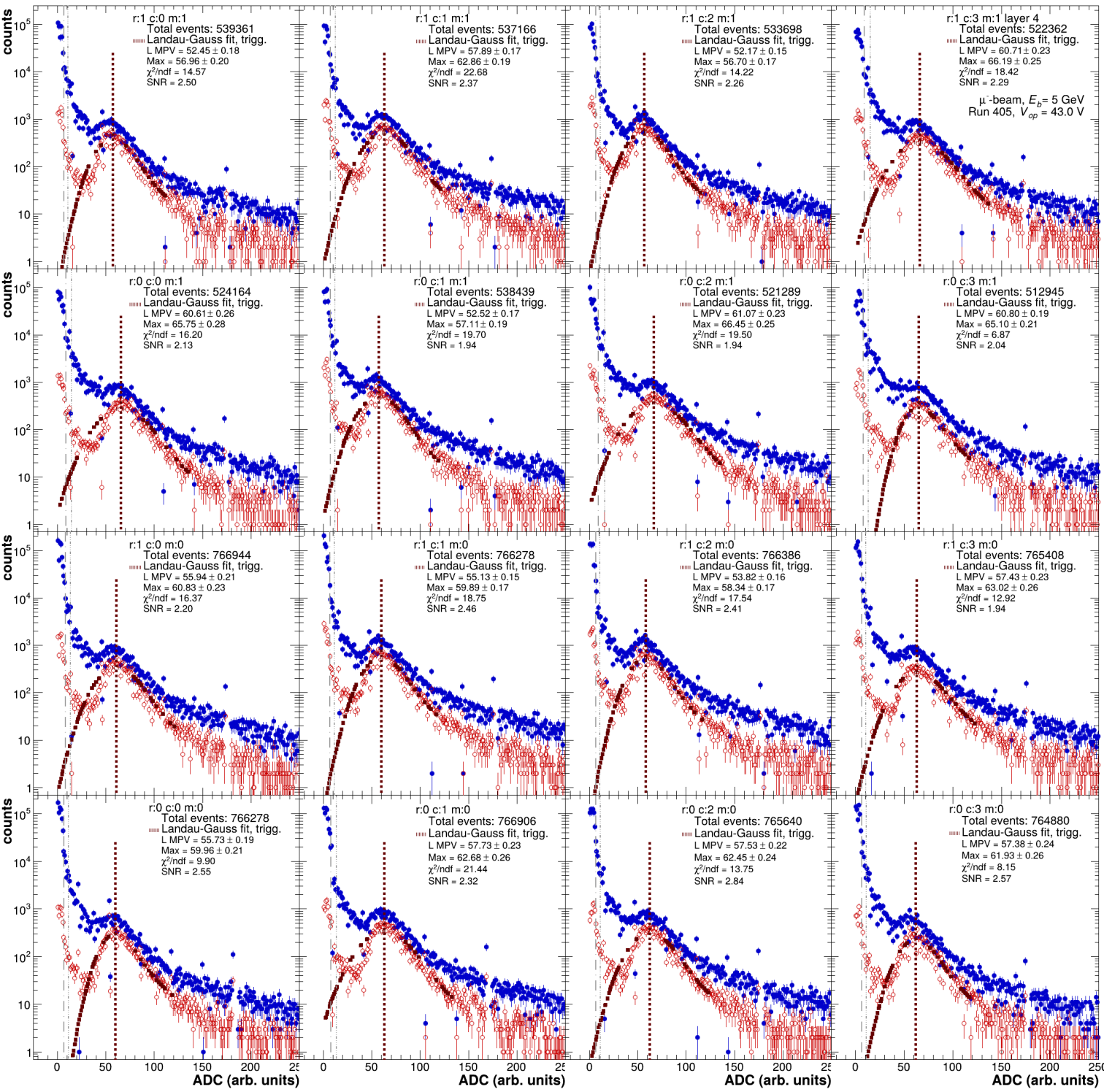

μ⁻-beam, E_b= 5 GeV, Run 405, V_{op} = 43.0 V
brd channel
layer
χ^2/ndf
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
5
10
15
20
25
30
35

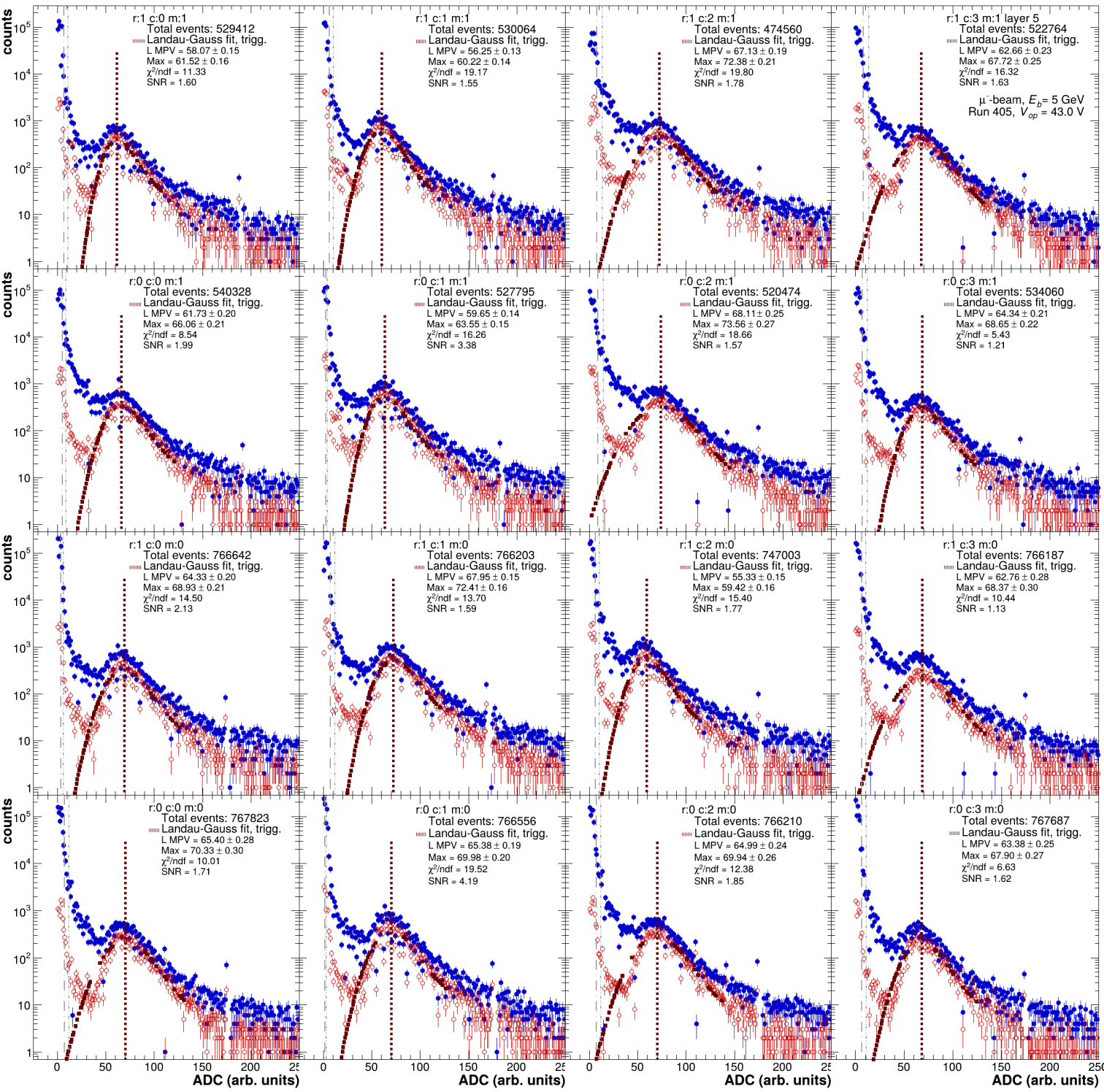

r:1 c:0 m:1
Total events: 514866
Landau-Gauss fit, trigg.
L MPV = 43.05 ± 0.13
Max = 46.80 ± 0.14
χ²/ndf = 13.82
SNR = 7.99
r:1 c:1 m:1
Total events: 476150
Landau-Gauss fit, trigg.
L MPV = 41.07 ± 0.09
Max = 44.25 ± 0.09
χ²/ndf = 24.74
SNR = 7.91
r:1 c:2 m:1
Total events: 432580
Landau-Gauss fit, trigg.
L MPV = 42.42 ± 0.10
Max = 46.09 ± 0.11
χ²/ndf = 37.23
SNR = 10.38
r:1 c:3 m:1 layer 0
Total events: 519124
Landau-Gauss fit, trigg.
L MPV = 42.00 ± 0.11
Max = 45.60 ± 0.12
χ²/ndf = 21.99
SNR = 7.54
μ⁻-beam, E_b = 5 GeV
Run 405, V_{op} = 43.0 V
r:0 c:0 m:1
Total events: 513618
Landau-Gauss fit, trigg.
L MPV = 42.16 ± 0.15
Max = 46.41 ± 0.17
χ²/ndf = 13.48
SNR = 4.35
r:0 c:1 m:1
Total events: 512393
Landau-Gauss fit, trigg.
L MPV = 37.97 ± 0.10
Max = 41.29 ± 0.11
χ²/ndf = 16.79
SNR = 3.63
r:0 c:2 m:1
Total events: 472911
Landau-Gauss fit, trigg.
L MPV = 40.34 ± 0.12
Max = 44.12 ± 0.13
χ²/ndf = 20.63
SNR = 4.06
r:0 c:3 m:1
Total events: 532673
Landau-Gauss fit, trigg.
L MPV = 35.54 ± 0.11
Max = 38.38 ± 0.12
χ²/ndf = 7.73
SNR = 2.79
r:1 c:0 m:0
Total events: 764506
Landau-Gauss fit, trigg.
L MPV = 28.48 ± 0.07
Max = 30.33 ± 0.08
χ²/ndf = 12.01
SNR = 5.29
r:1 c:1 m:0
Total events: 745683
Landau-Gauss fit, trigg.
L MPV = 31.32 ± 0.07
Max = 33.68 ± 0.07
χ²/ndf = 17.00
SNR = 7.12
r:1 c:2 m:0
Total events: 729182
Landau-Gauss fit, trigg.
L MPV = 28.50 ± 0.07
Max = 30.56 ± 0.07
χ²/ndf = 23.25
SNR = 5.34
r:1 c:3 m:0
Total events: 764299
Landau-Gauss fit, trigg.
L MPV = 29.59 ± 0.08
Max = 31.41 ± 0.08
χ²/ndf = 13.16
SNR = 8.24
r:0 c:0 m:0
Total events: 764908
Landau-Gauss fit, trigg.
L MPV = 29.36 ± 0.09
Max = 31.55 ± 0.10
χ²/ndf = 7.96
SNR = 6.10
r:0 c:1 m:0
Total events: 764242
Landau-Gauss fit, trigg.
L MPV = 30.73 ± 0.08
Max = 32.87 ± 0.08
χ²/ndf = 13.65
SNR = 5.92
r:0 c:2 m:0
Total events: 746066
Landau-Gauss fit, trigg.
L MPV = 28.74 ± 0.08
Max = 30.95 ± 0.09
χ²/ndf = 17.67
SNR = 7.06
r:0 c:3 m:0
counts
ADC (arb. units)

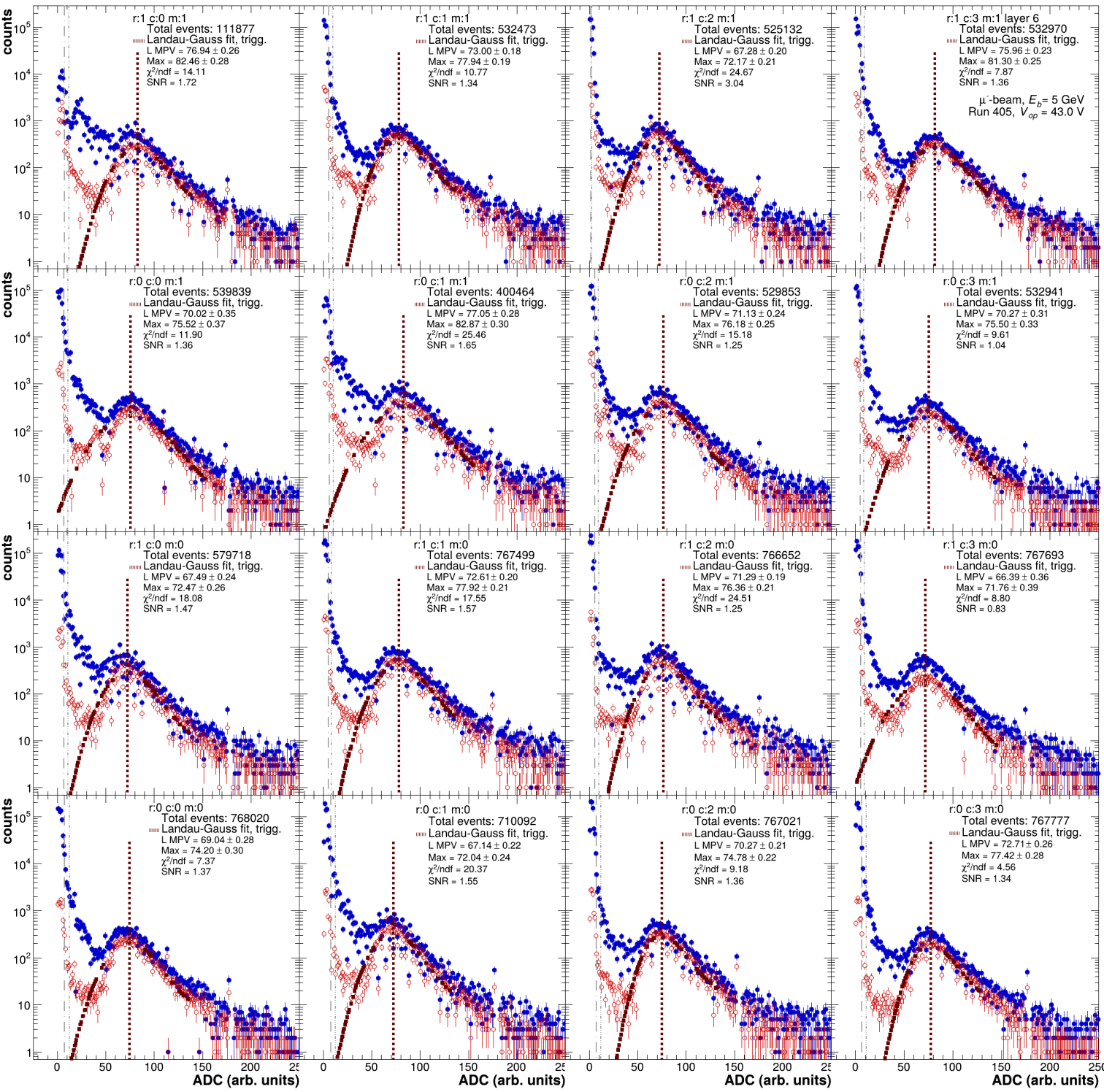

r:1 c:0 m:1
Total events: 512830
Landau-Gauss fit, trigg.
L MPV = 43.32 ± 0.13
Max = 47.33 ± 0.14
χ²/ndf = 14.48
SNR = 8.85
r:1 c:1 m:1
Total events: 513292
Landau-Gauss fit, trigg.
L MPV = 43.38 ± 0.10
Max = 47.17 ± 0.11
χ²/ndf = 17.10
SNR = 8.72
r:1 c:2 m:1
Total events: 514986
Landau-Gauss fit, trigg.
L MPV = 44.21 ± 0.10
Max = 48.11 ± 0.11
χ²/ndf = 26.17
SNR = 8.58
r:1 c:3 m:1 layer 1
Total events: 525586
Landau-Gauss fit, trigg.
L MPV = 41.01 ± 0.11
Max = 44.29 ± 0.11
χ²/ndf = 26.63
SNR = 9.15
μ⁻-beam, E_b = 5 GeV
Run 405, V_op = 43.0 V
r:0 c:0 m:1
Total events: 508610
Landau-Gauss fit, trigg.
L MPV = 41.56 ± 0.14
Max = 45.51 ± 0.15
χ²/ndf = 14.11
SNR = 6.61
r:0 c:1 m:1
Total events: 513543
Landau-Gauss fit, trigg.
L MPV = 44.18 ± 0.13
Max = 48.94 ± 0.15
χ²/ndf = 22.31
SNR = 5.06
r:0 c:2 m:1
Total events: 514988
Landau-Gauss fit, trigg.
L MPV = 42.73 ± 0.12
Max = 46.92 ± 0.13
χ²/ndf = 17.37
SNR = 5.11
r:0 c:3 m:1
Total events: 534565
Landau-Gauss fit, trigg.
L MPV = 43.10 ± 0.15
Max = 46.96 ± 0.16
χ²/ndf = 10.00
SNR = 3.90
r:1 c:0 m:0
Total events: 763588
Landau-Gauss fit, trigg.
L MPV = 30.09 ± 0.08
Max = 31.77 ± 0.09
χ²/ndf = 7.51
SNR = 7.22
r:1 c:1 m:0
Total events: 761936
Landau-Gauss fit, trigg.
L MPV = 31.92 ± 0.07
Max = 33.61 ± 0.07
χ²/ndf = 19.66
SNR = 7.13
r:1 c:2 m:0
Total events: 763003
Landau-Gauss fit, trigg.
L MPV = 28.17 ± 0.07
Max = 29.74 ± 0.07
χ²/ndf = 36.61
SNR = 6.59
r:1 c:3 m:0
Total events: 764162
Landau-Gauss fit, trigg.
L MPV = 29.91 ± 0.09
Max = 32.20 ± 0.10
χ²/ndf = 17.77
SNR = 11.80
r:0 c:0 m:0
Total events: 763458
Landau-Gauss fit, trigg.
L MPV = 31.42 ± 0.10
Max = 33.30 ± 0.10
χ²/ndf = 9.95
SNR = 8.13
r:0 c:1 m:0
Total events: 763596
Landau-Gauss fit, trigg.
L MPV = 33.25 ± 0.09
Max = 35.59 ± 0.09
χ²/ndf = 12.51
SNR = 8.96
r:0 c:2 m:0
Total events: 764300
Landau-Gauss fit, trigg.
L MPV = 30.09 ± 0.08
Max = 31.65 ± 0.09
χ²/ndf = 13.94
SNR = 6.90
r:0 c:3 m:0
Total events: 766405
Landau-Gauss fit, trigg.
L MPV = 30.52 ± 0.10
Max = 32.33 ± 0.11
χ²/ndf = 5.95
SNR = 5.73
counts
ADC (arb. units)

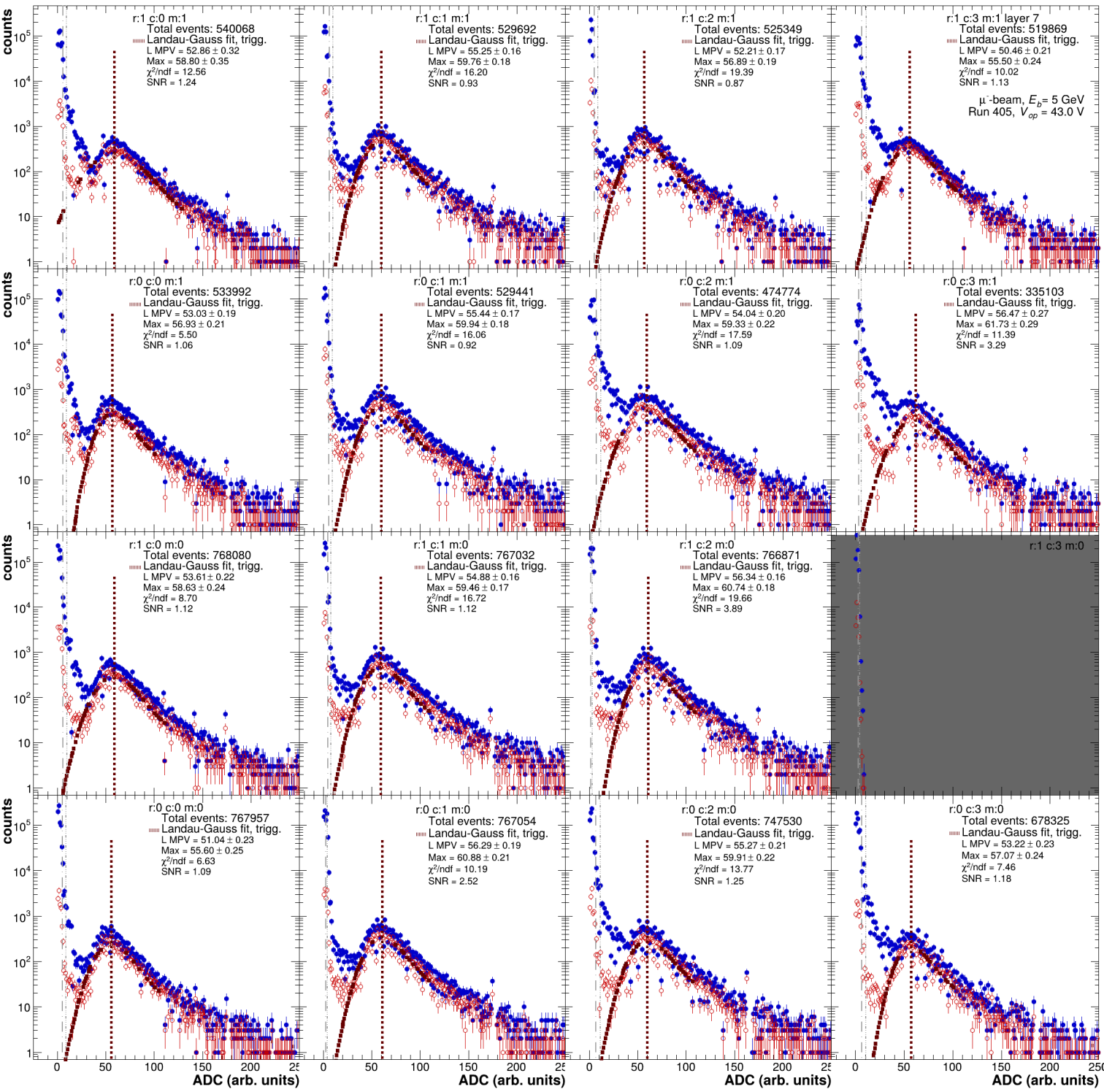

r:1 c:0 m:1
Total events: 478342
Landau-Gauss fit, trigg.
L MPV = 52.55 ± 0.19
Max = 57.70 ± 0.20
χ²/ndf = 22.77
SNR = 4.96
r:1 c:1 m:1
Total events: 531015
Landau-Gauss fit, trigg.
L MPV = 51.87 ± 0.18
Max = 57.76 ± 0.20
χ²/ndf = 28.40
SNR = 4.86
r:1 c:2 m:1
Total events: 471273
Landau-Gauss fit, trigg.
L MPV = 57.41 ± 0.19
Max = 63.43 ± 0.21
χ²/ndf = 36.34
SNR = 5.62
r:1 c:3 m:1 layer 2
Total events: 200320
Landau-Gauss fit, trigg.
L MPV = 51.97 ± 0.18
Max = 57.46 ± 0.20
χ²/ndf = 39.36
SNR = 6.18
μ⁻-beam, E_b = 5 GeV
Run 405, V_op = 43.0 V
r:0 c:0 m:1
Total events: 515083
Landau-Gauss fit, trigg.
L MPV = 53.00 ± 0.21
Max = 58.21 ± 0.23
χ²/ndf = 19.37
SNR = 4.02
r:0 c:1 m:1
Total events: 532740
Landau-Gauss fit, trigg.
L MPV = 51.07 ± 0.23
Max = 57.13 ± 0.26
χ²/ndf = 24.96
SNR = 4.00
r:0 c:2 m:1
Total events: 534141
Landau-Gauss fit, trigg.
L MPV = 47.28 ± 0.19
Max = 52.77 ± 0.22
χ²/ndf = 17.03
SNR = 3.73
r:0 c:3 m:1
Total events: 535523
Landau-Gauss fit, trigg.
L MPV = 54.58 ± 0.27
Max = 60.57 ± 0.30
χ²/ndf = 14.85
SNR = 3.80
r:1 c:0 m:0
Total events: 746141
Landau-Gauss fit, trigg.
L MPV = 54.28 ± 0.22
Max = 59.91 ± 0.25
χ²/ndf = 19.81
SNR = 4.75
r:1 c:1 m:0
Total events: 763814
Landau-Gauss fit, trigg.
L MPV = 49.83 ± 0.15
Max = 54.83 ± 0.17
χ²/ndf = 27.25
SNR = 5.13
r:1 c:2 m:0
Total events: 743242
Landau-Gauss fit, trigg.
L MPV = 51.89 ± 0.18
Max = 57.27 ± 0.20
χ²/ndf = 23.99
SNR = 4.59
r:1 c:3 m:0
Total events: 612863
Landau-Gauss fit, trigg.
L MPV = 48.88 ± 0.36
Max = 56.08 ± 0.41
χ²/ndf = 17.19
SNR = 3.05
r:0 c:0 m:0
Total events: 764322
Landau-Gauss fit, trigg.
L MPV = 52.82 ± 0.25
Max = 58.19 ± 0.28
χ²/ndf = 12.87
SNR = 4.85
r:0 c:1 m:0
Total events: 765282
Landau-Gauss fit, trigg.
L MPV = 52.62 ± 0.19
Max = 57.56 ± 0.21
χ²/ndf = 17.27
SNR = 4.56
r:0 c:2 m:0
Total events: 765912
Landau-Gauss fit, trigg.
L MPV = 49.36 ± 0.19
Max = 54.00 ± 0.21
χ²/ndf = 16.65
SNR = 4.20
r:0 c:3 m:0
Total events: 766331
Landau-Gauss fit, trigg.
L MPV = 52.18 ± 0.26
Max = 57.32 ± 0.29
χ²/ndf = 11.82
SNR = 4.82
counts
ADC (arb. units)

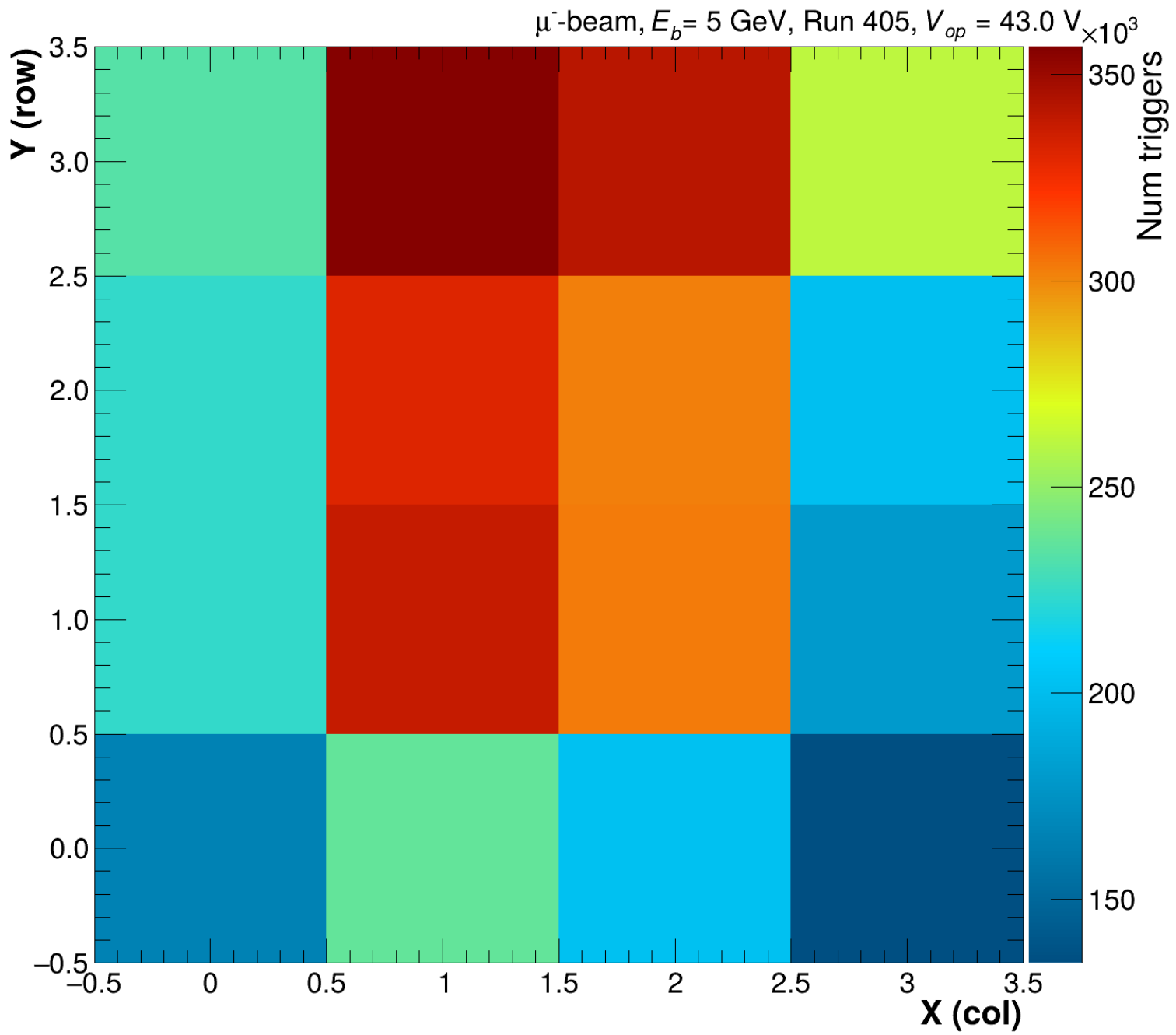

r:1 c:0 m:1
Total events: 527749
Landau-Gauss fit, trigg.
L MPV = 54.38 ± 0.25
Max = 59.48 ± 0.27
χ²/ndf = 14.74
SNR = 0.82
r:1 c:1 m:1
Total events: 285235
Landau-Gauss fit, trigg.
L MPV = 56.32 ± 0.25
Max = 62.31 ± 0.28
χ²/ndf = 29.21
SNR = 1.59
r:1 c:2 m:1
Total events: 533805
Landau-Gauss fit, trigg.
L MPV = 56.60 ± 0.25
Max = 62.39 ± 0.27
χ²/ndf = 16.16
SNR = 0.61
r:1 c:3 m:1 layer 3
Total events: 540053
Landau-Gauss fit, trigg.
L MPV = 52.99 ± 0.46
Max = 59.40 ± 0.51
χ²/ndf = 21.05
SNR = 0.66
μ⁻-beam, E_b = 5 GeV
Run 405, V_op = 43.0 V
r:0 c:0 m:1
Total events: 529441
Landau-Gauss fit, trigg.
L MPV = 53.57 ± 0.27
Max = 58.65 ± 0.29
χ²/ndf = 13.67
SNR = 0.51
r:0 c:1 m:1
Total events: 518241
Landau-Gauss fit, trigg.
L MPV = 54.74 ± 0.34
Max = 61.35 ± 0.38
χ²/ndf = 12.70
SNR = 0.66
r:0 c:2 m:1
Total events: 474213
Landau-Gauss fit, trigg.
L MPV = 52.50 ± 0.32
Max = 58.88 ± 0.35
χ²/ndf = 15.67
SNR = 0.65
r:0 c:3 m:1
Total events: 532742
Landau-Gauss fit, trigg.
L MPV = 59.29 ± 0.35
Max = 64.97 ± 0.39
χ²/ndf = 7.97
SNR = 0.50
r:1 c:0 m:0
Total events: 767093
Landau-Gauss fit, trigg.
L MPV = 56.37 ± 0.26
Max = 61.71 ± 0.29
χ²/ndf = 18.26
SNR = 2.12
r:1 c:1 m:0
Total events: 656927
Landau-Gauss fit, trigg.
L MPV = 55.26 ± 0.23
Max = 60.77 ± 0.25
χ²/ndf = 36.52
SNR = 1.29
r:1 c:2 m:0
Total events: 767826
Landau-Gauss fit, trigg.
L MPV = 56.42 ± 0.23
Max = 61.89 ± 0.26
χ²/ndf = 12.77
SNR = 0.60
r:1 c:3 m:0
r:0 c:0 m:0
Total events: 767189
Landau-Gauss fit, trigg.
L MPV = 51.89 ± 0.28
Max = 56.85 ± 0.30
χ²/ndf = 11.08
SNR = 3.35
r:0 c:1 m:0
Total events: 766520
Landau-Gauss fit, trigg.
L MPV = 57.69 ± 0.32
Max = 63.62 ± 0.35
χ²/ndf = 10.47
SNR = 0.86
r:0 c:2 m:0
Total events: 747545
Landau-Gauss fit, trigg.
L MPV = 56.46 ± 0.31
Max = 62.34 ± 0.34
χ²/ndf = 11.60
SNR = 1.20
r:0 c:3 m:0
Total events: 767905
Landau-Gauss fit, trigg.
L MPV = 54.57 ± 0.33
Max = 60.07 ± 0.37
χ²/ndf = 6.13
SNR = 0.58
counts
ADC (arb. units)

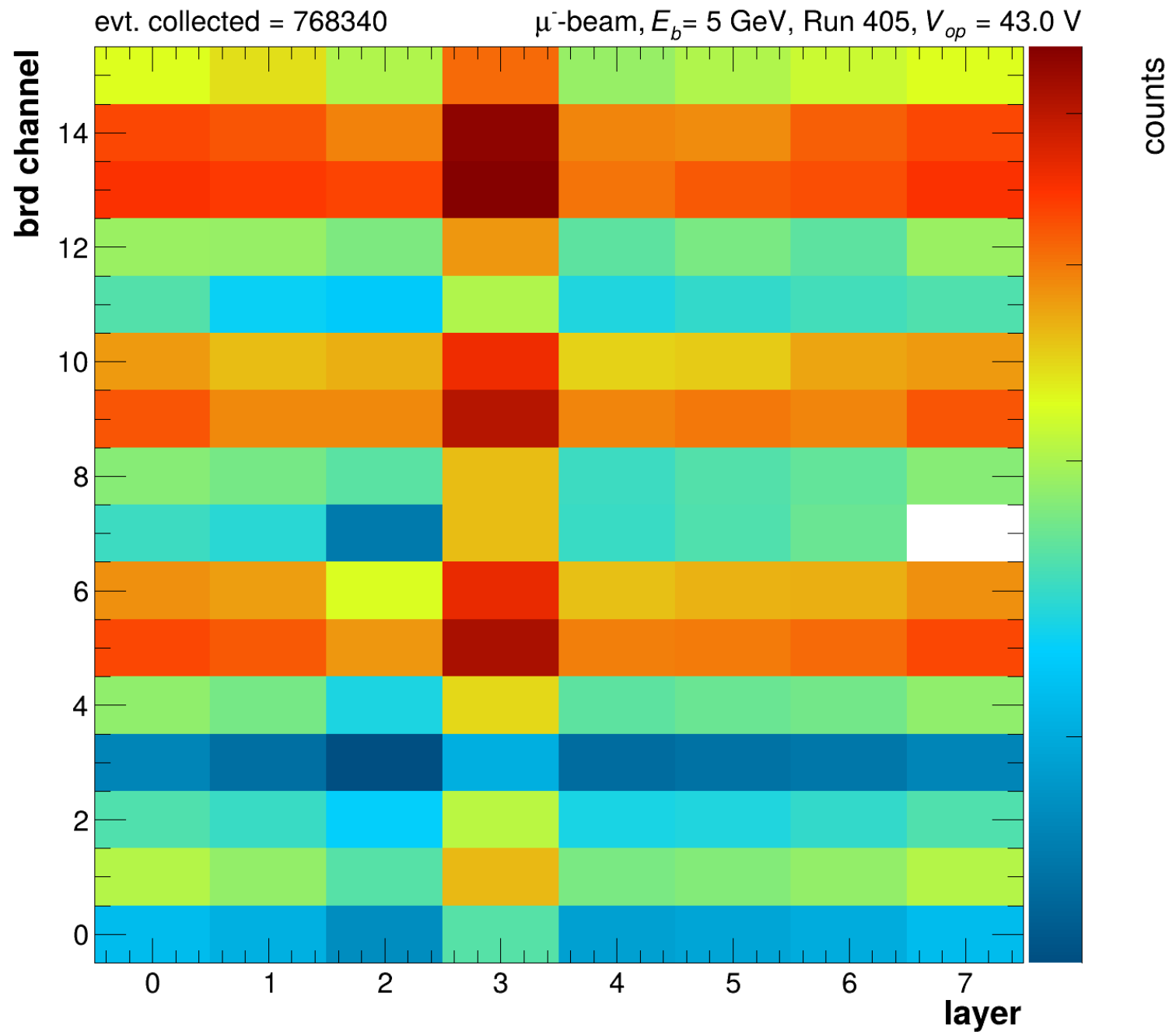

r:1 c:0 m:1
Total events: 539361
Landau-Gauss fit, trigg.
L MPV = 52.45 ± 0.18
Max = 56.96 ± 0.20
χ²/ndf = 14.57
SNR = 2.50
r:1 c:1 m:1
Total events: 537166
Landau-Gauss fit, trigg.
L MPV = 57.89 ± 0.17
Max = 62.86 ± 0.19
χ²/ndf = 22.68
SNR = 2.37
r:1 c:2 m:1
Total events: 533698
Landau-Gauss fit, trigg.
L MPV = 52.17 ± 0.15
Max = 56.70 ± 0.17
χ²/ndf = 14.22
SNR = 2.26
r:1 c:3 m:1 layer 4
Total events: 522362
Landau-Gauss fit, trigg.
L MPV = 60.71 ± 0.23
Max = 66.19 ± 0.25
χ²/ndf = 18.42
SNR = 2.29
μ⁻-beam, E_b = 5 GeV
Run 405, V_op = 43.0 V
r:0 c:0 m:1
Total events: 524164
Landau-Gauss fit, trigg.
L MPV = 60.61 ± 0.26
Max = 65.75 ± 0.28
χ²/ndf = 16.20
SNR = 2.13
r:0 c:1 m:1
Total events: 538439
Landau-Gauss fit, trigg.
L MPV = 52.52 ± 0.17
Max = 57.11 ± 0.19
χ²/ndf = 19.70
SNR = 1.94
r:0 c:2 m:1
Total events: 521289
Landau-Gauss fit, trigg.
L MPV = 61.07 ± 0.23
Max = 66.45 ± 0.25
χ²/ndf = 19.50
SNR = 1.94
r:0 c:3 m:1
Total events: 512945
Landau-Gauss fit, trigg.
L MPV = 60.80 ± 0.19
Max = 65.10 ± 0.21
χ²/ndf = 6.87
SNR = 2.04
r:1 c:0 m:0
Total events: 766944
Landau-Gauss fit, trigg.
L MPV = 55.94 ± 0.21
Max = 60.83 ± 0.23
χ²/ndf = 16.37
SNR = 2.20
r:1 c:1 m:0
Total events: 766278
Landau-Gauss fit, trigg.
L MPV = 55.13 ± 0.15
Max = 59.89 ± 0.17
χ²/ndf = 18.75
SNR = 2.46
r:1 c:2 m:0
Total events: 766386
Landau-Gauss fit, trigg.
L MPV = 53.82 ± 0.16
Max = 58.34 ± 0.17
χ²/ndf = 17.54
SNR = 2.41
r:1 c:3 m:0
Total events: 765408
Landau-Gauss fit, trigg.
L MPV = 57.43 ± 0.23
Max = 63.02 ± 0.26
χ²/ndf = 12.92
SNR = 1.94
r:0 c:0 m:0
Total events: 766278
Landau-Gauss fit, trigg.
L MPV = 55.73 ± 0.19
Max = 59.96 ± 0.21
χ²/ndf = 9.90
SNR = 2.55
r:0 c:1 m:0
Total events: 766906
Landau-Gauss fit, trigg.
L MPV = 57.73 ± 0.23
Max = 62.68 ± 0.26
χ²/ndf = 21.44
SNR = 2.32
r:0 c:2 m:0
Total events: 765640
Landau-Gauss fit, trigg.
L MPV = 57.53 ± 0.22
Max = 62.45 ± 0.24
χ²/ndf = 13.75
SNR = 2.84
r:0 c:3 m:0
Total events: 764880
Landau-Gauss fit, trigg.
L MPV = 57.38 ± 0.24
Max = 61.93 ± 0.26
χ²/ndf = 8.15
SNR = 2.57
counts
ADC (arb. units)

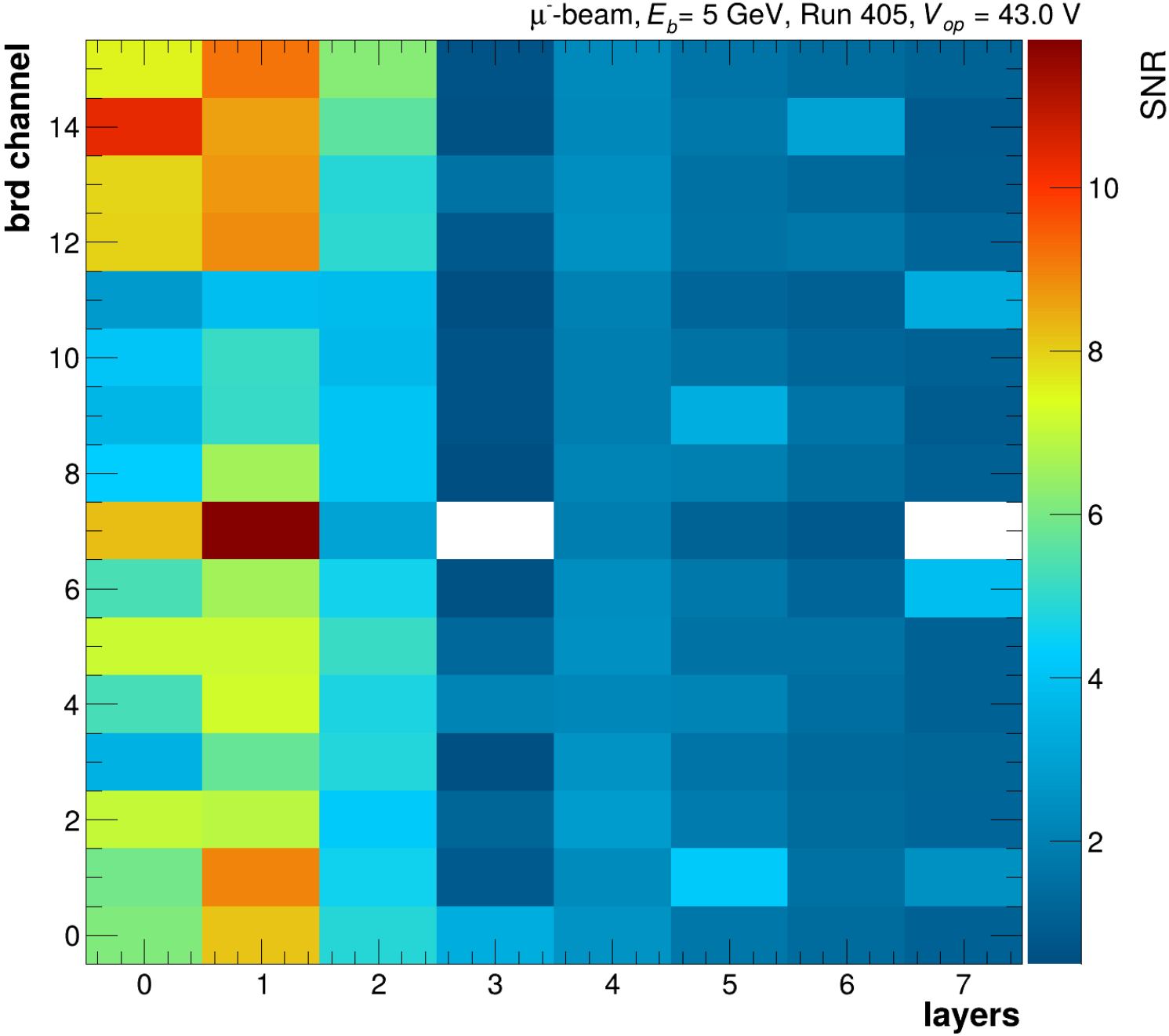

r:1 c:0 m:1
Total events: 529412
Landau-Gauss fit, trigg.
L MPV = 58.07 ± 0.15
Max = 61.52 ± 0.16
χ²/ndf = 11.33
SNR = 1.60
r:1 c:1 m:1
Total events: 530064
Landau-Gauss fit, trigg.
L MPV = 56.25 ± 0.13
Max = 60.22 ± 0.14
χ²/ndf = 19.17
SNR = 1.55
r:1 c:2 m:1
Total events: 474560
Landau-Gauss fit, trigg.
L MPV = 67.13 ± 0.19
Max = 72.38 ± 0.21
χ²/ndf = 19.80
SNR = 1.78
r:1 c:3 m:1 layer 5
Total events: 522764
Landau-Gauss fit, trigg.
L MPV = 62.66 ± 0.23
Max = 67.72 ± 0.25
χ²/ndf = 16.32
SNR = 1.63
μ⁻-beam, E_b = 5 GeV
Run 405, V_op = 43.0 V
r:0 c:0 m:1
Total events: 540328
Landau-Gauss fit, trigg.
L MPV = 61.73 ± 0.20
Max = 66.06 ± 0.21
χ²/ndf = 8.54
SNR = 1.99
r:0 c:1 m:1
Total events: 527795
Landau-Gauss fit, trigg.
L MPV = 59.65 ± 0.14
Max = 63.55 ± 0.15
χ²/ndf = 16.26
SNR = 3.38
r:0 c:2 m:1
Total events: 520474
Landau-Gauss fit, trigg.
L MPV = 68.11 ± 0.25
Max = 73.56 ± 0.27
χ²/ndf = 18.66
SNR = 1.57
r:0 c:3 m:1
Total events: 534060
Landau-Gauss fit, trigg.
L MPV = 64.34 ± 0.21
Max = 68.65 ± 0.22
χ²/ndf = 5.43
SNR = 1.21
r:1 c:0 m:0
Total events: 766642
Landau-Gauss fit, trigg.
L MPV = 64.33 ± 0.20
Max = 68.93 ± 0.21
χ²/ndf = 14.50
SNR = 2.13
r:1 c:1 m:0
Total events: 766203
Landau-Gauss fit, trigg.
L MPV = 67.95 ± 0.15
Max = 72.41 ± 0.16
χ²/ndf = 13.70
SNR = 1.59
r:1 c:2 m:0
Total events: 747003
Landau-Gauss fit, trigg.
L MPV = 55.33 ± 0.15
Max = 59.42 ± 0.16
χ²/ndf = 15.40
SNR = 1.77
r:1 c:3 m:0
Total events: 766187
Landau-Gauss fit, trigg.
L MPV = 62.76 ± 0.28
Max = 68.37 ± 0.30
χ²/ndf = 10.44
SNR = 1.13
r:0 c:0 m:0
Total events: 767823
Landau-Gauss fit, trigg.
L MPV = 65.40 ± 0.28
Max = 70.33 ± 0.30
χ²/ndf = 10.01
SNR = 1.71
r:0 c:1 m:0
Total events: 766556
Landau-Gauss fit, trigg.
L MPV = 65.38 ± 0.19
Max = 69.98 ± 0.20
χ²/ndf = 19.52
SNR = 4.19
r:0 c:2 m:0
Total events: 766210
Landau-Gauss fit, trigg.
L MPV = 64.99 ± 0.24
Max = 69.94 ± 0.26
χ²/ndf = 12.38
SNR = 1.85
r:0 c:3 m:0
Total events: 767687
Landau-Gauss fit, trigg.
L MPV = 63.38 ± 0.25
Max = 67.90 ± 0.27
χ²/ndf = 6.63
SNR = 1.62
counts
ADC (arb. units)

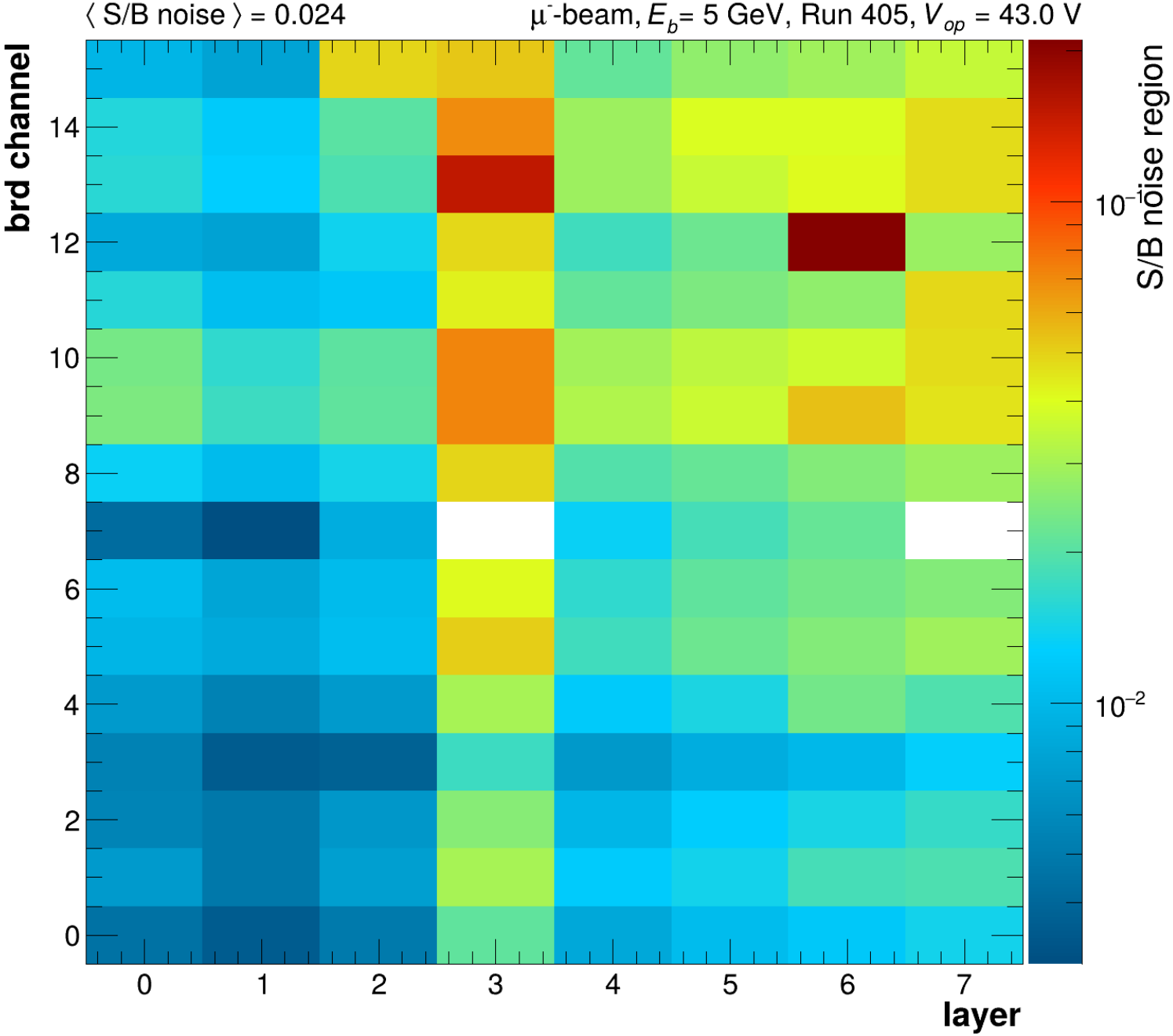

r:1 c:0 m:1
Total events: 111877
Landau-Gauss fit, trigg.
L MPV = 76.94 ± 0.26
Max = 82.46 ± 0.28
χ²/ndf = 14.11
SNR = 1.72
r:1 c:1 m:1
Total events: 532473
Landau-Gauss fit, trigg.
L MPV = 73.00 ± 0.18
Max = 77.94 ± 0.19
χ²/ndf = 10.77
SNR = 1.34
r:1 c:2 m:1
Total events: 525132
Landau-Gauss fit, trigg.
L MPV = 67.28 ± 0.20
Max = 72.17 ± 0.21
χ²/ndf = 24.67
SNR = 3.04
r:1 c:3 m:1 layer 6
Total events: 532970
Landau-Gauss fit, trigg.
L MPV = 75.96 ± 0.23
Max = 81.30 ± 0.25
χ²/ndf = 7.87
SNR = 1.36
μ⁻-beam, E_b = 5 GeV
Run 405, V_op = 43.0 V
r:0 c:0 m:1
Total events: 539839
Landau-Gauss fit, trigg.
L MPV = 70.02 ± 0.35
Max = 75.52 ± 0.37
χ²/ndf = 11.90
SNR = 1.36
r:0 c:1 m:1
Total events: 400464
Landau-Gauss fit, trigg.
L MPV = 77.05 ± 0.28
Max = 82.87 ± 0.30
χ²/ndf = 25.46
SNR = 1.65
r:0 c:2 m:1
Total events: 529853
Landau-Gauss fit, trigg.
L MPV = 71.13 ± 0.24
Max = 76.18 ± 0.25
χ²/ndf = 15.18
SNR = 1.25
r:0 c:3 m:1
Total events: 532941
Landau-Gauss fit, trigg.
L MPV = 70.27 ± 0.31
Max = 75.50 ± 0.33
χ²/ndf = 9.61
SNR = 1.04
r:1 c:0 m:0
Total events: 579718
Landau-Gauss fit, trigg.
L MPV = 67.49 ± 0.24
Max = 72.47 ± 0.26
χ²/ndf = 18.08
SNR = 1.47
r:1 c:1 m:0
Total events: 767499
Landau-Gauss fit, trigg.
L MPV = 72.61 ± 0.20
Max = 77.92 ± 0.21
χ²/ndf = 17.55
SNR = 1.57
r:1 c:2 m:0
Total events: 766652
Landau-Gauss fit, trigg.
L MPV = 71.29 ± 0.19
Max = 76.36 ± 0.21
χ²/ndf = 24.51
SNR = 1.25
r:1 c:3 m:0
Total events: 767693
Landau-Gauss fit, trigg.
L MPV = 66.39 ± 0.36
Max = 71.76 ± 0.39
χ²/ndf = 8.80
SNR = 0.83
r:0 c:0 m:0
Total events: 768020
Landau-Gauss fit, trigg.
L MPV = 69.04 ± 0.28
Max = 74.20 ± 0.30
χ²/ndf = 7.37
SNR = 1.37
r:0 c:1 m:0
Total events: 710092
Landau-Gauss fit, trigg.
L MPV = 67.14 ± 0.22
Max = 72.04 ± 0.24
χ²/ndf = 20.37
SNR = 1.55
r:0 c:2 m:0
Total events: 767021
Landau-Gauss fit, trigg.
L MPV = 70.27 ± 0.21
Max = 74.78 ± 0.22
χ²/ndf = 9.18
SNR = 1.36
r:0 c:3 m:0
Total events: 767777
Landau-Gauss fit, trigg.
L MPV = 72.71 ± 0.26
Max = 77.42 ± 0.28
χ²/ndf = 4.56
SNR = 1.34
counts
ADC (arb. units)

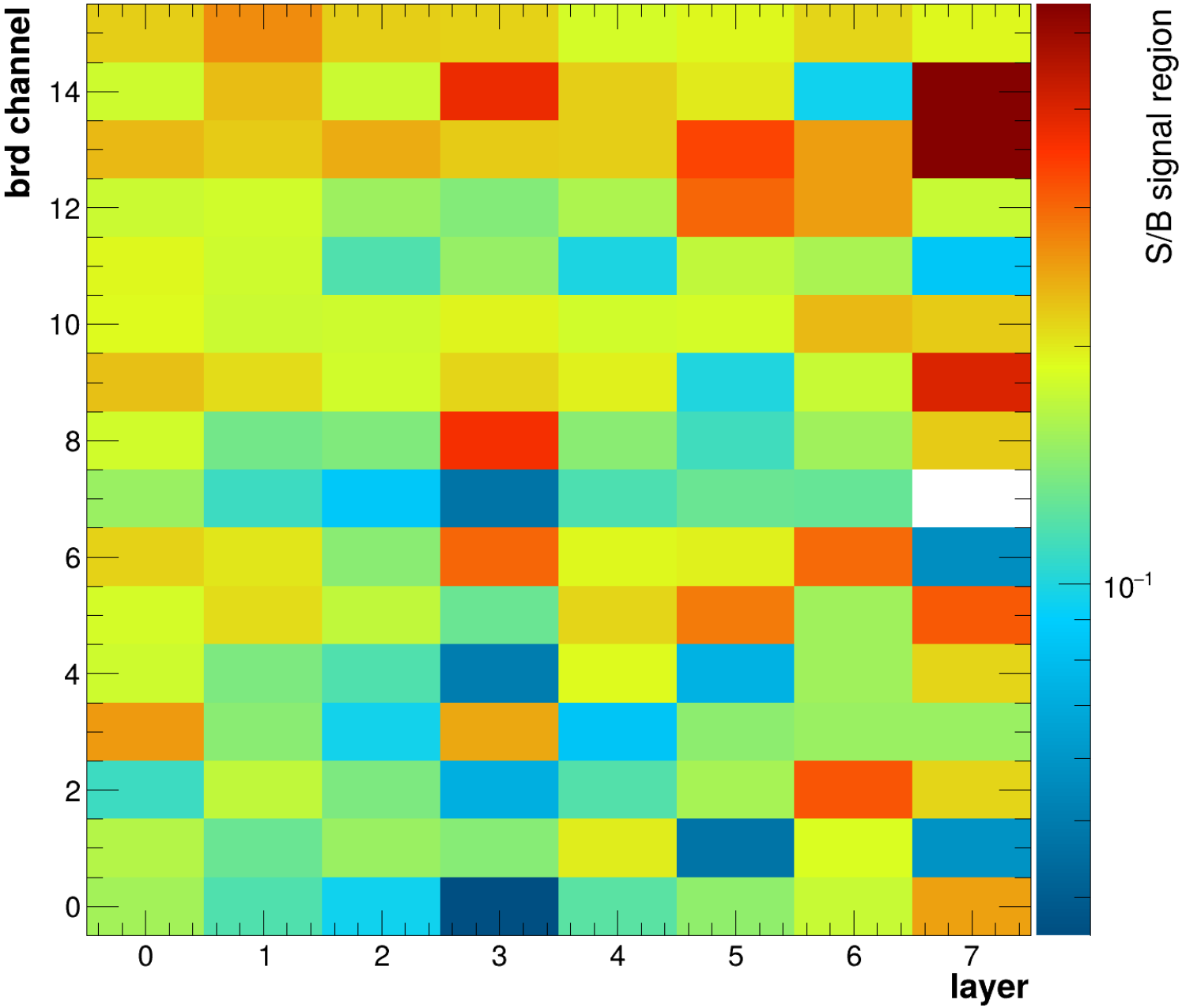

r:1 c:0 m:1
Total events: 540068
Landau-Gauss fit, trigg.
L MPV = 52.86 ± 0.32
Max = 58.80 ± 0.35
χ²/ndf = 12.56
SNR = 1.24
r:1 c:1 m:1
Total events: 529692
Landau-Gauss fit, trigg.
L MPV = 55.25 ± 0.16
Max = 59.76 ± 0.18
χ²/ndf = 16.20
SNR = 0.93
r:1 c:2 m:1
Total events: 525349
Landau-Gauss fit, trigg.
L MPV = 52.21 ± 0.17
Max = 56.89 ± 0.19
χ²/ndf = 19.39
SNR = 0.87
r:1 c:3 m:1 layer 7
Total events: 519869
Landau-Gauss fit, trigg.
L MPV = 50.46 ± 0.21
Max = 55.50 ± 0.24
χ²/ndf = 10.02
SNR = 1.13
μ⁻-beam, E_b = 5 GeV
Run 405, V_op = 43.0 V
r:0 c:0 m:1
Total events: 533992
Landau-Gauss fit, trigg.
L MPV = 53.03 ± 0.19
Max = 56.93 ± 0.21
χ²/ndf = 5.50
SNR = 1.06
r:0 c:1 m:1
Total events: 529441
Landau-Gauss fit, trigg.
L MPV = 55.44 ± 0.17
Max = 59.94 ± 0.18
χ²/ndf = 16.06
SNR = 0.92
r:0 c:2 m:1
Total events: 474774
Landau-Gauss fit, trigg.
L MPV = 54.04 ± 0.20
Max = 59.33 ± 0.22
χ²/ndf = 17.59
SNR = 1.09
r:0 c:3 m:1
Total events: 335103
Landau-Gauss fit, trigg.
L MPV = 56.47 ± 0.27
Max = 61.73 ± 0.29
χ²/ndf = 11.39
SNR = 3.29
r:1 c:0 m:0
Total events: 768080
Landau-Gauss fit, trigg.
L MPV = 53.61 ± 0.22
Max = 58.63 ± 0.24
χ²/ndf = 8.70
SNR = 1.12
r:1 c:1 m:0
Total events: 767032
Landau-Gauss fit, trigg.
L MPV = 54.88 ± 0.16
Max = 59.46 ± 0.17
χ²/ndf = 16.72
SNR = 1.12
r:1 c:2 m:0
Total events: 766871
Landau-Gauss fit, trigg.
L MPV = 56.34 ± 0.16
Max = 60.74 ± 0.18
χ²/ndf = 19.66
SNR = 3.89
r:1 c:3 m:0
r:0 c:0 m:0
Total events: 767957
Landau-Gauss fit, trigg.
L MPV = 51.04 ± 0.23
Max = 55.60 ± 0.25
χ²/ndf = 6.63
SNR = 1.09
r:0 c:1 m:0
Total events: 767054
Landau-Gauss fit, trigg.
L MPV = 56.29 ± 0.19
Max = 60.88 ± 0.21
χ²/ndf = 10.19
SNR = 2.52
r:0 c:2 m:0
Total events: 747530
Landau-Gauss fit, trigg.
L MPV = 55.27 ± 0.21
Max = 59.91 ± 0.22
χ²/ndf = 13.77
SNR = 1.25
r:0 c:3 m:0
Total events: 678325
Landau-Gauss fit, trigg.
L MPV = 53.22 ± 0.23
Max = 57.07 ± 0.24
χ²/ndf = 7.46
SNR = 1.18
counts
ADC (arb. units)

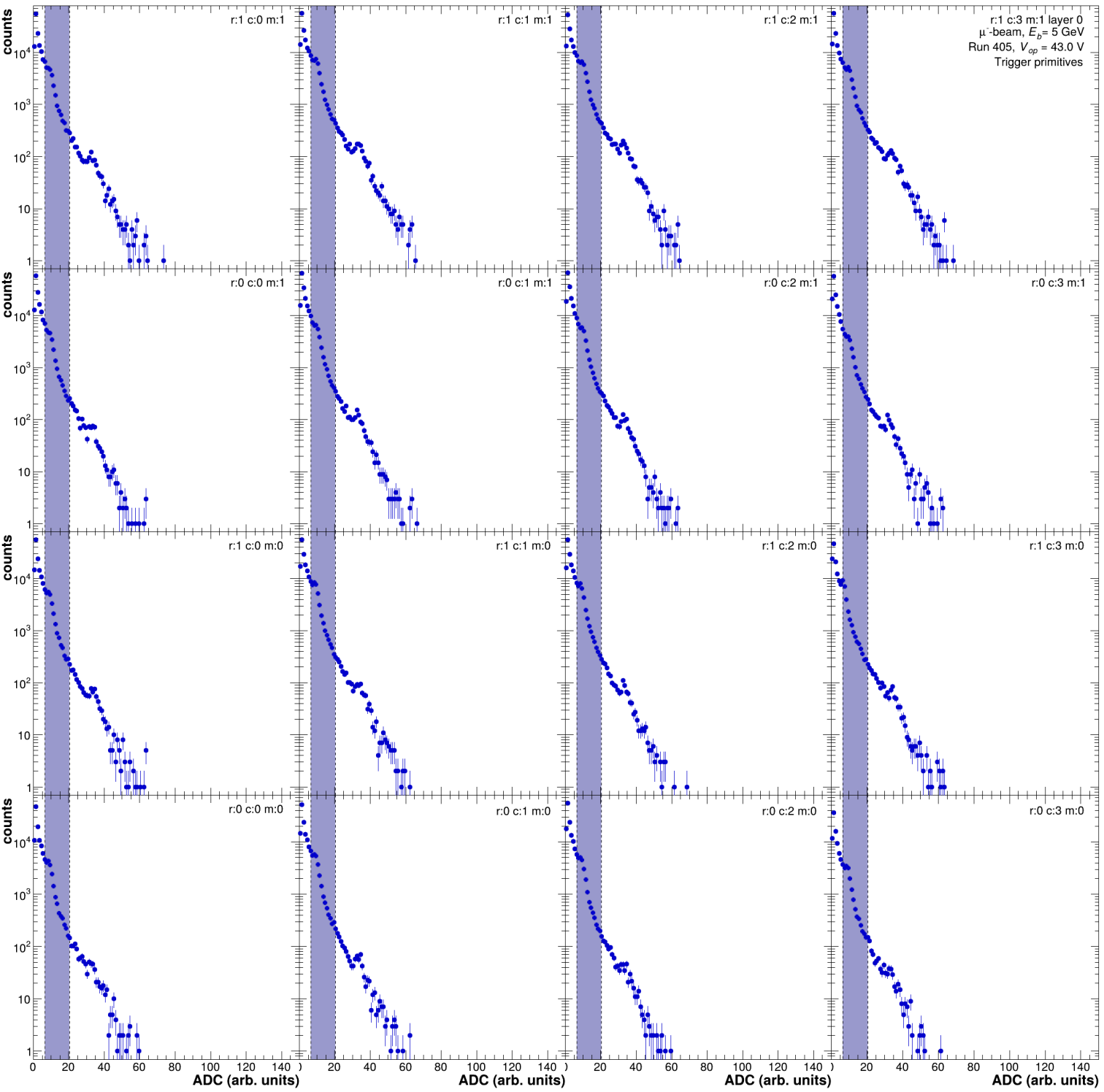

μ⁻-beam, E_b= 5 GeV, Run 405, V_{op} = 43.0 V
Y (row)
X (col)
Num triggers
×10³
350
300
250
200
150
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−0.5
0
0.5
1
1.5
2
2.5
3
3.5

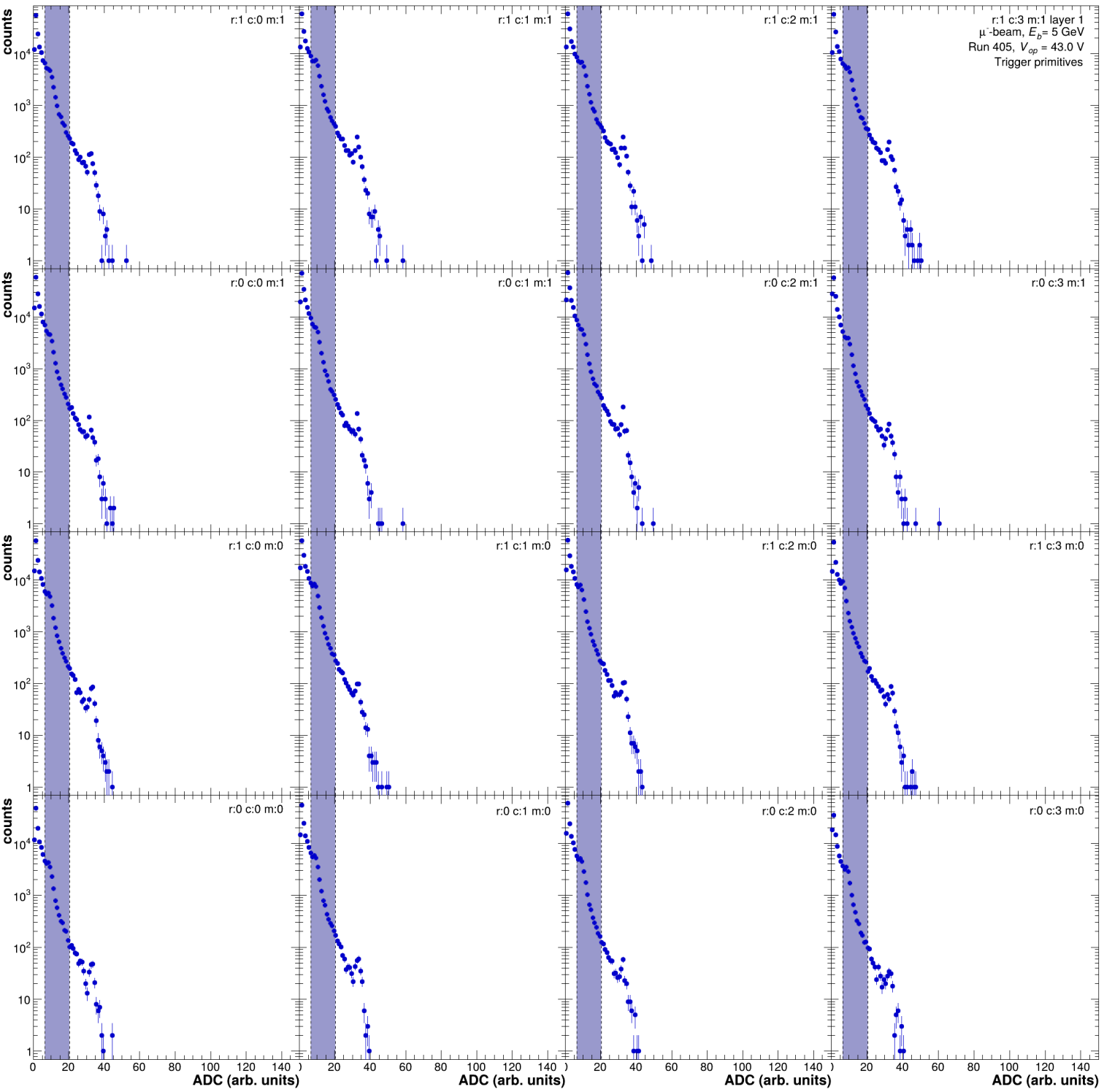

evt. collected = 768340
μ⁻-beam, E_b= 5 GeV, Run 405, V_{op} = 43.0 V
brd channel
14
12
10
8
6
4
2
0
0
1
2
3
4
5
6
7
layer
counts

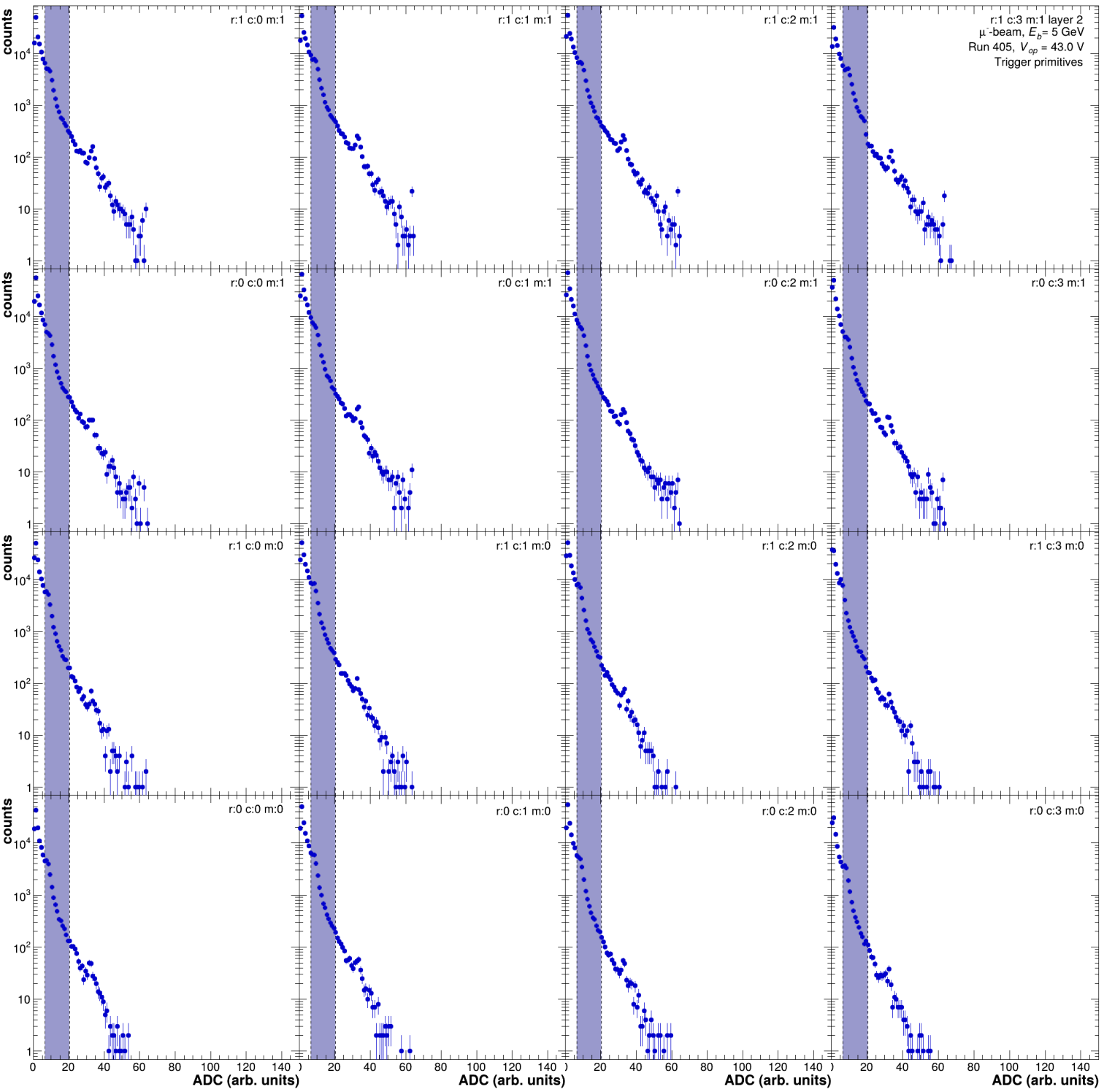

μ⁻-beam, E_b= 5 GeV, Run 405, V_{op} = 43.0 V
brd channel
0
2
4
6
8
10
12
14
layers
0
1
2
3
4
5
6
7
SNR
2
4
6
8
10

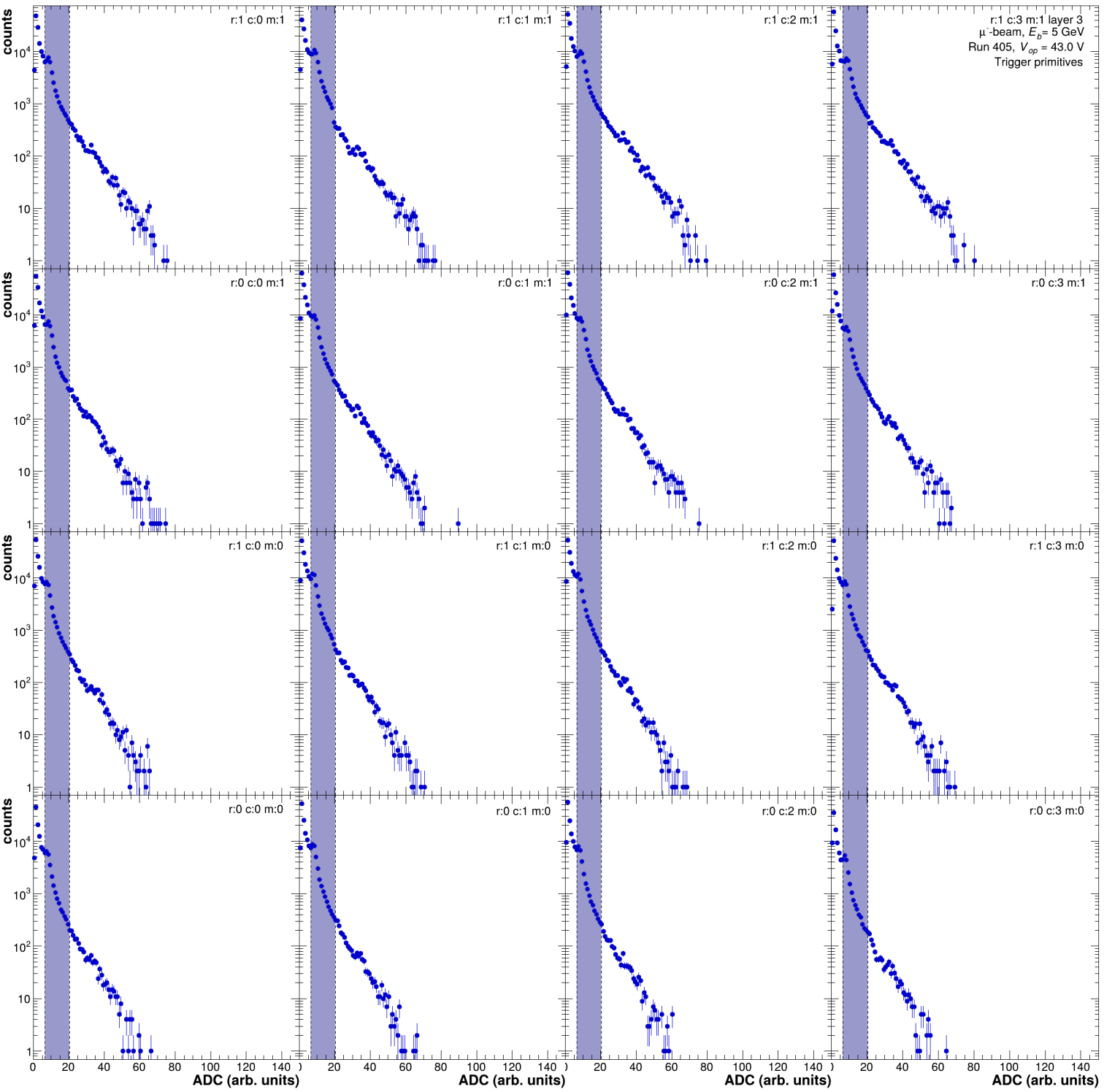

⟨ S/B noise ⟩ = 0.024
μ⁻-beam, E_b= 5 GeV, Run 405, V_{op} = 43.0 V
brd channel
layer
S/B noise region
10^{-1}
10^{-2}
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7

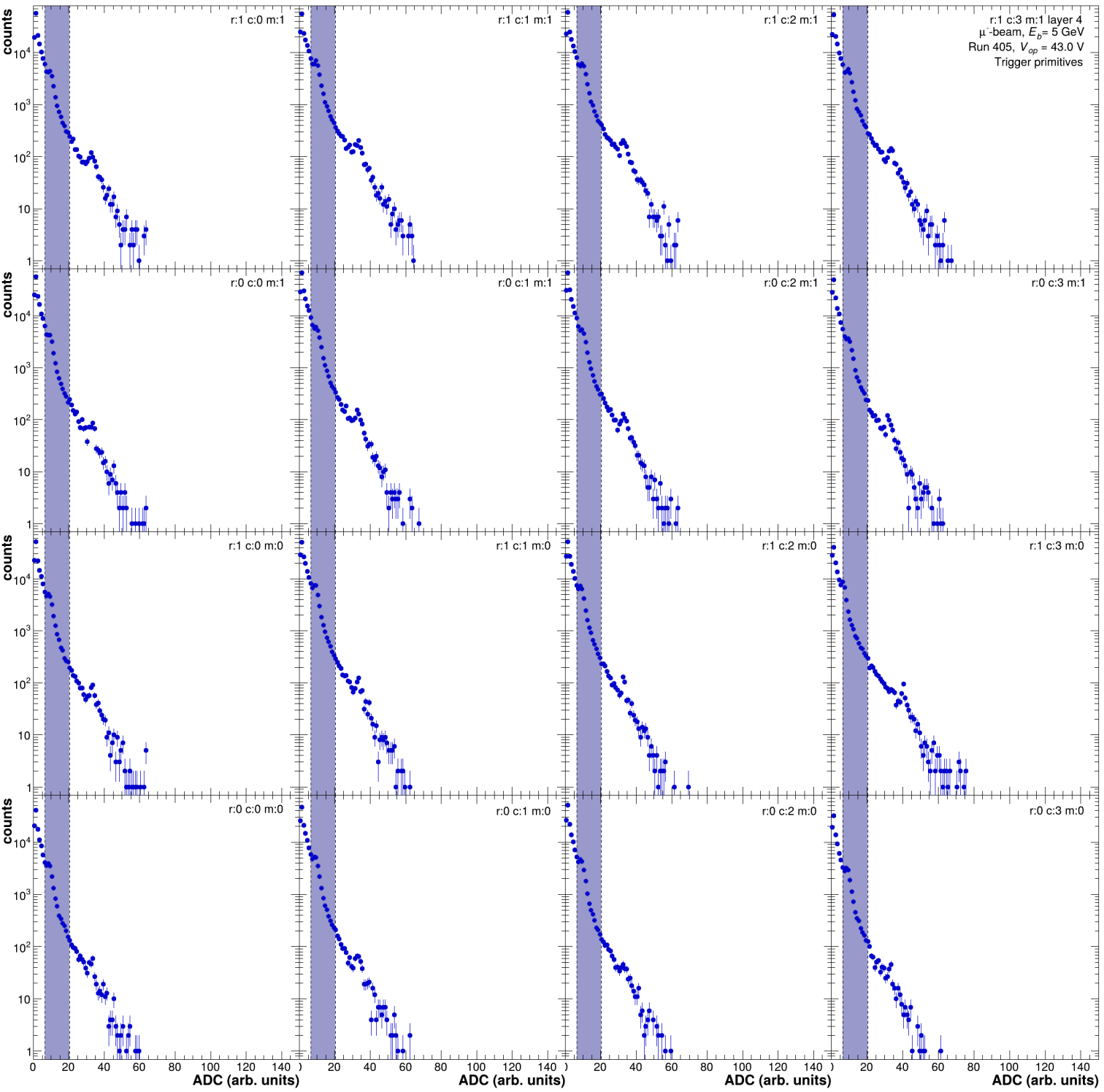

⟨ S/B signal ⟩ = 0.182
μ⁻-beam, E_b= 5 GeV, Run 405, V_{op} = 43.0 V
brd channel
layer
S/B signal region
10^{-1}
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7

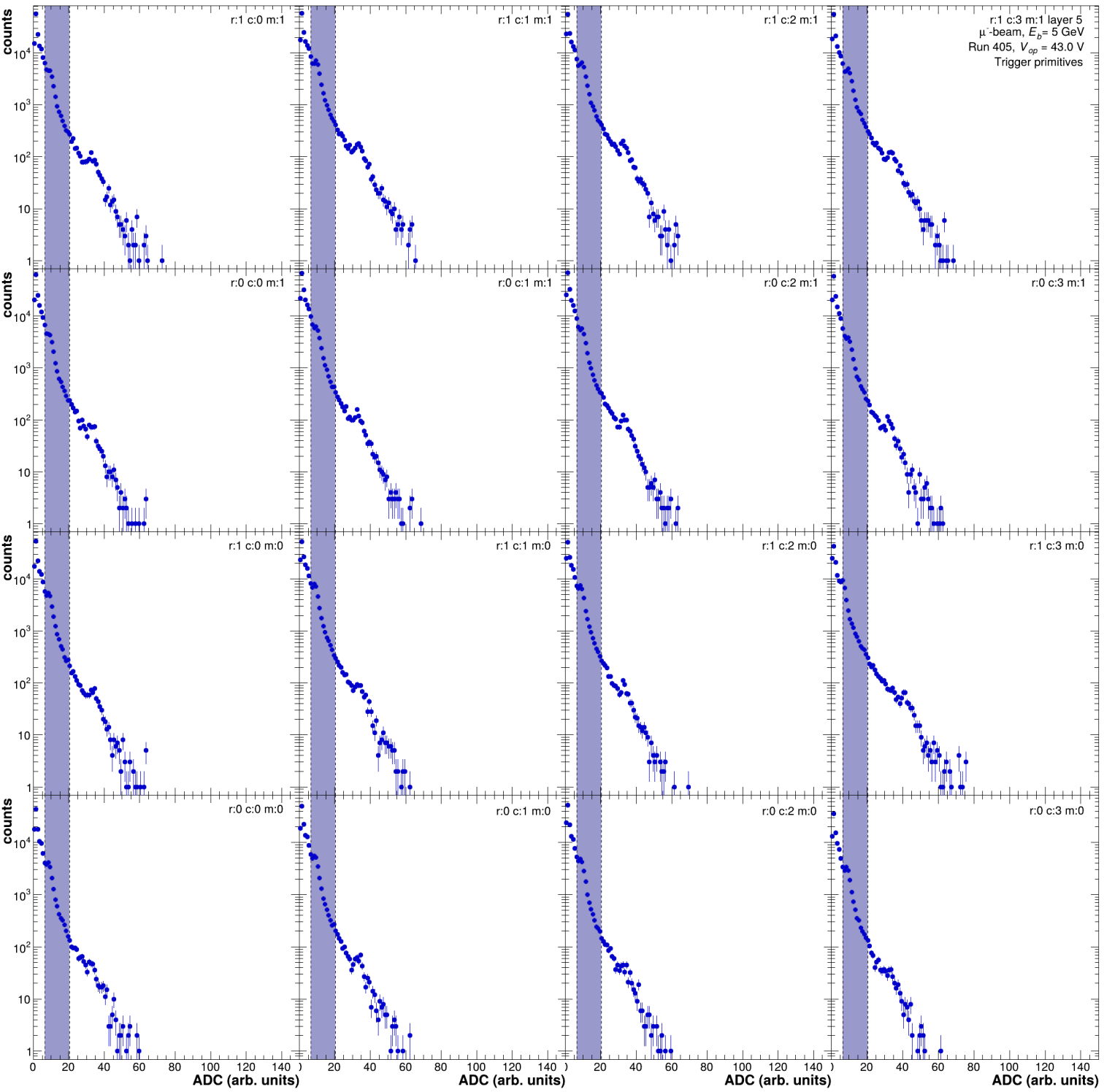

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 0
μ⁻-beam, E_b= 5 GeV
Run 405, V_{op} = 43.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)

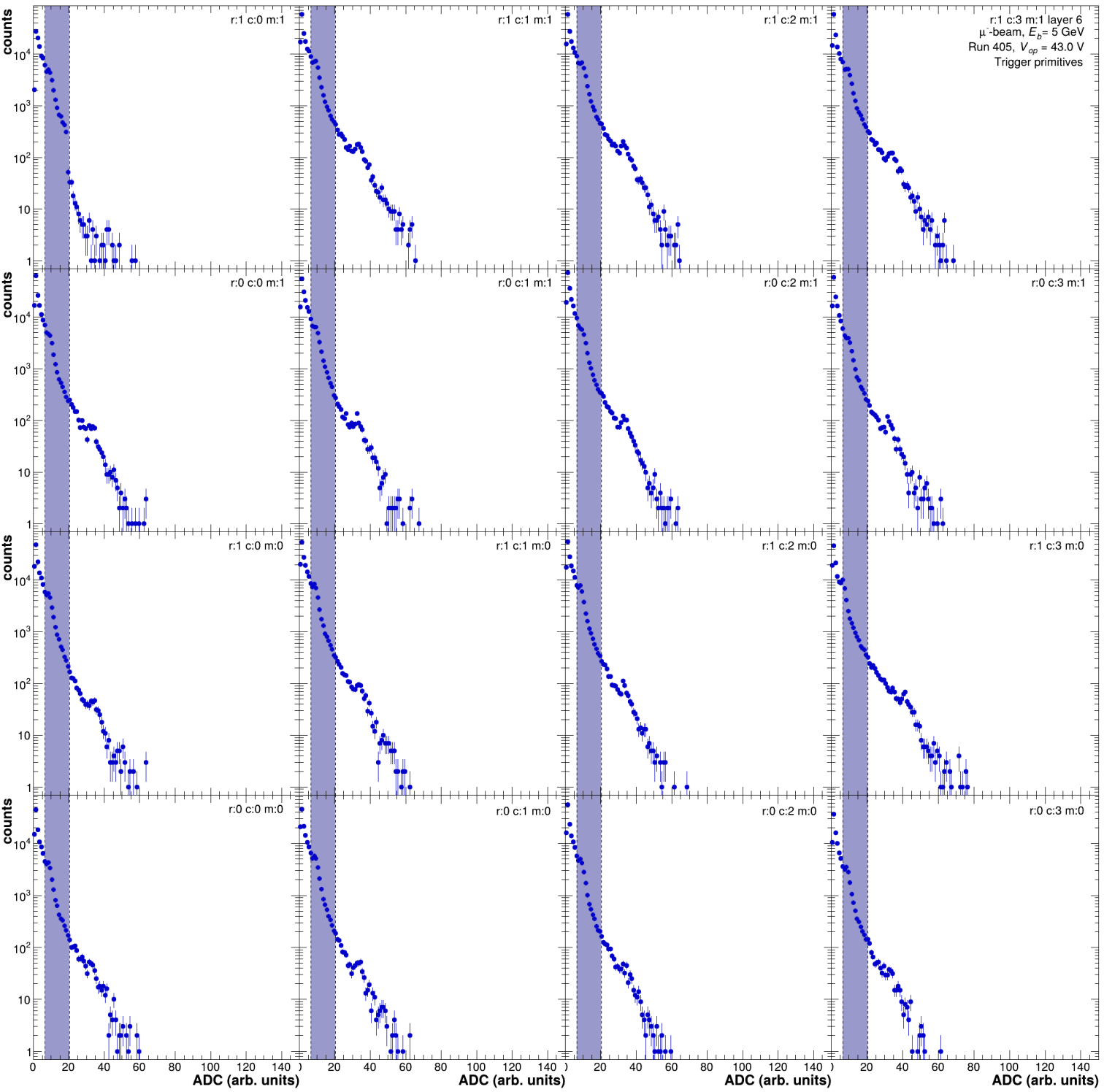

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 1
μ⁻-beam, E_b= 5 GeV
Run 405, V_{op} = 43.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)

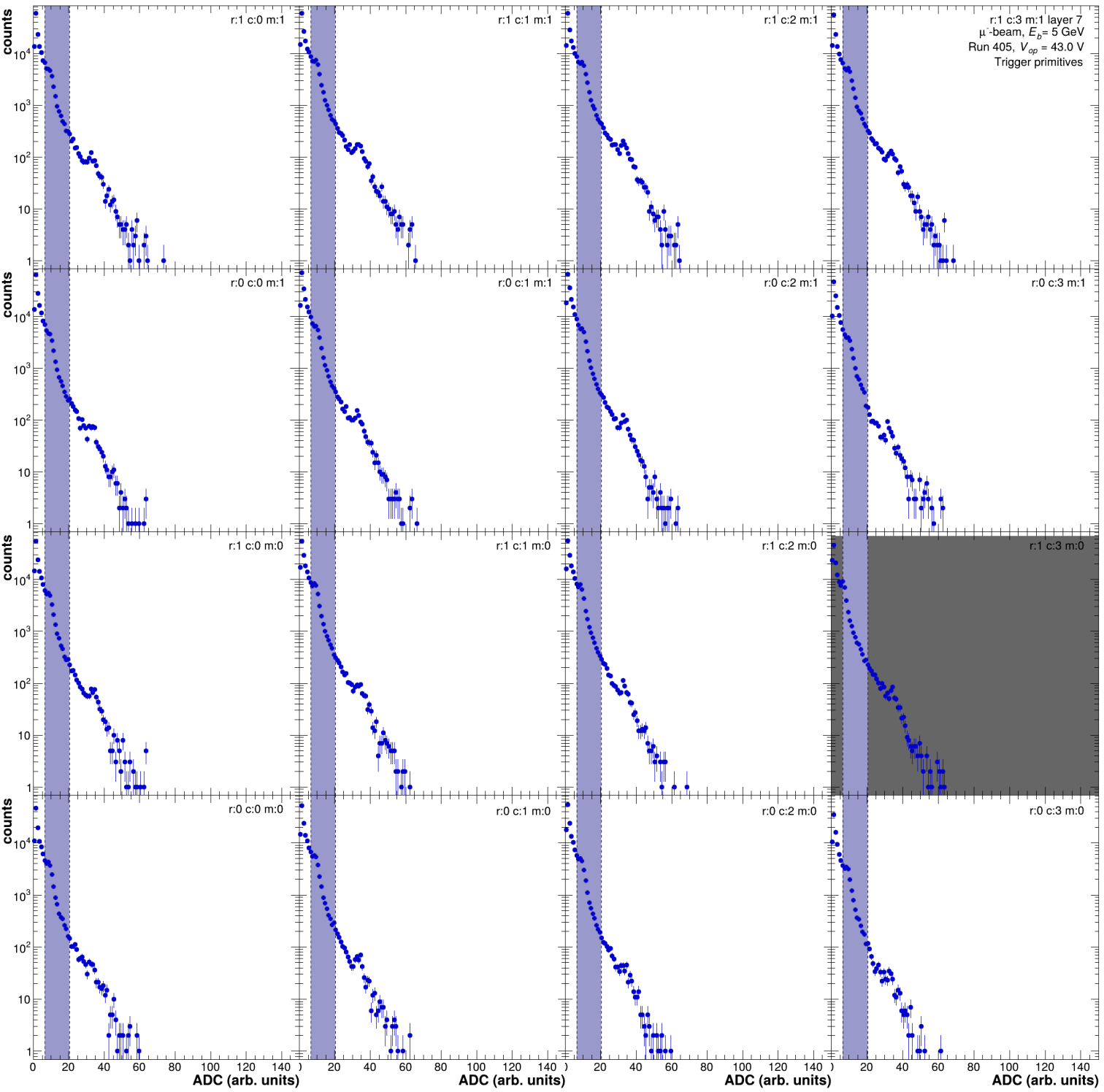

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 2
μ⁻-beam, E_b= 5 GeV
Run 405, V_{op} = 43.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 3
μ⁻-beam, E_b= 5 GeV
Run 405, V_{op} = 43.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 4
μ⁻-beam, E_b= 5 GeV
Run 405, V_{op} = 43.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 5
μ⁻-beam, E_b= 5 GeV
Run 405, V_{op} = 43.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 6
μ⁻-beam, E_b= 5 GeV
Run 405, V_{op} = 43.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 7
μ⁻-beam, E_b= 5 GeV
Run 405, V_{op} = 43.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)